Polarization Controlled Microwave Fields for Atomic Spin Ground State Manipulation

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Polarisationskontrollierte Mikrowellenfelder zur Spinmanipulation des atomaren Grundzustandes

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Abstract

The coherent manipulation of hyperfine ground states in alkali-metal atoms by means of microwave magnetic fields is well established in experimental atomic physics. The microwave magnetic field interacts with the ground state magnetic dipole moment, resulting in a coupling between hyperfine Zeeman states, with a coupling strength that depends on the microwave field polarization. The underlying theory that describes the coupling between hyperfine Zeeman states of ^{87}Rb $5S_{1/2}$ manifold is elaborated and experimental methods which measure, manipulate and determine the microwave field polarization are described and experimentally studied on single atoms trapped in a double high-finesse cavity setup. Before performing the experiment on the atoms, a separate test apparatus is used for independent preliminary studies. Here, a microwave field polarization is measured and aligned at a spatial point of interest with an average fidelity of $(98.9 \pm 0.5\ )\%$. To measure the microwave field polarization with a single atom, microwave spectra, as well as the measurement of Rabi oscillations for certain ground state transitions, are taken. The adjustment of certain microwave field polarizations qualitatively follows the theoretical expectations. Underlying imperfections are discussed, and methods for improvements are presented.

Contents

1. Introduction

The beginning of the 20^{th} century made a profound mark on the history of physics and on science in general: on 14^{th} December 1900 Max Planck presented his work to the German Physical Society, where he presented his black-body radiation law, which assumes that electromagnetic energy can be emitted and absorbed only in discrete form, in this way solving the "ultraviolet catastrophe" [1]. Nowadays, this assumption is regarded as the birth of *Quantum Mechanics* and he was awarded the Nobel Prize in Physics in 1918 for his contribution to this new branch of physics.
This new theory had a huge impact on the scientific world, calling into question our knowledge of fundamental laws of nature and leading us to readjust the ideas of classical mechanics into a new, more profound picture of physics, based on *quanta*.
During the same century numerous discoveries and experimental verifications of quantum mechanics followed; among those, Einstein's early interpretation of the photoelectric effect [2] based on Hertz's experimental results [3], de Broglie's particle-wave duality [4] proved for the first time by Jönsson [5], Heisenberg's uncertainty principle [6], and many others [7, 8, 9, 10]. Nowadays it is an established theory and its accurate predictive power allows us to work with single atoms and photons and to probe their quantum features in everyday laboratory life.
Besides challenging the 20^{th} century's scientific community with increasingly unexpected aspects of nature, quantum mechanics has presented a significant technological impact since its advent. Consider computers for instance, whose performance has been made possible by the understanding of semiconductor constituents, which are correctly modeled as quantum systems, leading to the creation of field-effect transistors in 1947 [11], or the invention of lasers in 1960 [12]. The development of increasingly advanced technologies and the diffusion of some of these in everyday life has led to the digital revolution, whose revolutionary principle is the representation of information in a discrete manner, namely in *bits* - the fundamental units of classical information. Potential technological applications have increased over time and more recently the scientific and industrial interest in quantum computation has exploded. The unit of quantum information is called *qubit* and is known as the quantum counterpart of a classical bit; any two-level quantum system can represent a qubit, e.g. two electronic states of a single atom or the polarization states of a single photon.
The research in this direction has been inspired by the striking capabilities that quantum systems have if exploited as computational resources; the possibility of being able to perform computation at velocities never seen before and solving computational tasks otherwise impossible to accomplish is the driving force of this new technological revolution, aimed at reaching the so-called *quantum supremacy*. Evidence of this international interest are the European Union's flagship program, set up in 2018 and investing 1 billion Euros for the development of quantum technologies and more recently published research in this field [13].
It is essential in the branch of quantum computation, as indeed for classical computation, to be able to store units of information and to *process* them - naturally

in a quantum fashion. Rabi oscillations in such a context are a key element: in the presence of an oscillatory driving field, a two-level system will cycle between its two eigenstates; when an atom is physically impinged by a coherent beam of photons, it will coherently absorb photons and re-emit them by stimulated emission. The ability to control such oscillations between energy eigenstates of an atom allows the control over the atomic state, which can be put in an arbitrary superposition of these two states: this is a key resource for exploiting the potential of quantum systems for information processing.

As they are innate quantum objects, atoms constitute a good system for encoding qubits, where very often two electronic states play the role of qubit states. In particular, single neutral atoms coupled to high-finesse optical resonators have been shown to be a powerful platform for the implementation of many protocols for long-distance quantum information processing [14, 15, 16]. Photon qubits can travel long distances before meeting the addressed atom, and the interaction between the two quantum systems is mediated by the optical resonator (cavity), whose coupling to the atom enhances the atom-field interaction, thus enabling strong light-matter coupling.

Two common tools for manipulating atomic qubits in cavities are Raman [17] and microwave transitions. Microwaves (MW) can be tuned to the ground state hyperfine transitions in alkali-metal atoms and therefore allow a direct state transfer, as opposed to Raman transitions. The most commonly used tools for emitting MW fields in quantum optics experiments in vacuum at present are antennas. There are many types of antennas, with different shapes, dimensions and geometries but all of them have in common that the apparatus's geometry and near-by materials complicate predictions of the field intensity and the polarization in a certain point of the space, e.g. the atom's location. This thesis aims for the employment of polarization controlled MW fields in order to manipulate the electronic ground state of a single atom coupled to high-finesse cavities.

Experimental setups employing MW electrodes [18] on single alkali ions trapped in chips [19] have achieved a polarization control with error 2×10^{-5} in $^{43}Ca^{+}$ "atomic clock" qubits. The coherent manipulation of MW fields ensured their cancellation even in delimited zones, allowing a spatial MW addressing of qubits separated by almost 1 mm. Such setups are nowadays one of the most promising quantum computing platforms [20], particularly when hyperfine ground clock-state qubits are employed. However, they are not suitable for neutral atoms.

As for many experiments in quantum optics, ^{87}Rb atoms were used for this thesis, and qubit states were chosen among the manifold $5S_{1/2}$, since these are coupled dominantly by magnetic-dipole interaction and therefore are long-lived. The manifold $5S_{1/2}$ is affected by the hyperfine splitting, and therefore it is possible to identify more states, which can be coupled by MW fields, since in ^{87}Rb the hyperfine ground state splitting is about $\sim 6.8\,\mathrm{GHz}$ [21]. The setup consists of two crossed high-finesse fiber cavities where the atoms are trapped in their modes' crossing point. This thesis is focused on measuring, controlling and predicting the polarization of MW fields generated by the interference of three single-loop antennas, by means of MW spectroscopy and Rabi oscillation measurements. The successful MW manipulation in such systems could open the way for new protocols, or improve the performance of previous ones.

Being able to have control over the MW field polarization, and therefore to select a defined polarization, offers several advantages, among those the ability to choose

defined atomic transitions and suppress undesired ones - e.g. allowing direct population transfer in decoherence-protected states [14] and suppressing AC Zeeman shifts (AC for alternating current) in the ground state hyperfine transitions of $5S_{1/2}$ - or, if only one particular polarization is required, to avoid power dispersion in other polarizations.
The thesis is structured as follows:
In Chapter 2, the interaction between MW fields and the atom is treated and theoretical derivations of Rabi frequencies involving the atomic structure of ^{87}Rb are introduced. The theory concerning the polarization in two dimensions (2D) and three dimensions (3D) and the constructive interference of magnetic fields is discussed. Additionally, the AC Zeeman shift on the ground state clock transition due to the interaction between the ground state magnetic dipole moment and the MW magnetic field is estimated and the suppression of certain polarizations leading to a vanishing shift is demonstrated. However, an experimental confirmation is pending.
In Chapter 3, a MW antennas test setup for measuring and controlling MW fields with loop antennas as emitters and dipolar antennas as receivers is shown, and a strategy for obtaining a defined polarization and related results are discussed. This apparatus is independent of the main cavity setup, where atoms are trapped, and is located outside the chamber.
In Chapter 4, the cavities' and MW setup in the vacuum chamber are presented: fiber cavities, state preparation, MW control system and state detection are detailed. As for the test apparatus, the implementation of a strategy for measuring and predicting defined MW polarizations is examined. The related results are reported, analyzed and discussed; MW spectra before and after MW polarization manipulation show only partial suppression of the undesired polarizations. The major limiting factors are discussed. Due to the appearance of an issue in the experimental setup, described in the Appendix, the aims of this project could not be fully achieved, revealing nevertheless room for improvement.
In Chapter 5, a summary of the main results, together with an overview of future improvements, concludes this thesis.
The Appendix outlines the issues in the short cavity encountered during the last stages of the project. The inclusion of this chapter is motivated by the fact that future experiments could benefit from an accurate report on the issues encountered and offer some ideas and strategies that might be helpful to tackle them.

2. Theory

2.1. Microwave-atom interaction

The problem of describing the transition of the state population in an idealized two-level system due to the interaction with an harmonic coherent field takes the name of Rabi Problem, named after Isidor Isaac Rabi, who found the solution in the context of magnetic resonance [22]. Since this concept is the key of the physics adopted for this work, it is worth to shortly recap the main results.
Here, the case of an oscillating magnetic field applied on a two-level atom is discussed. The treatment of the interaction between atoms and fields in such a frame is frequently called "semiclassical" because the atom is treated as a quantum object, whereas the field is completely classical.
If $\mathbf{B}(t)$ describes the magnetic field and $\boldsymbol{\mu}$ represents the atomic magnetic dipole moment, the Hamiltonian describing the atomic interaction with the magnetic field is

$$\hat{H}_B(t) = -\boldsymbol{\mu} \cdot \mathbf{B}(t) = \frac{\mu_B}{\hbar}(g_S\mathbf{S} + g_I\mathbf{I} + g_L\mathbf{L}) \cdot \mathbf{B}(t) \tag{2.1}$$

In this Hamiltonian, $\mathbf{S}$, $\mathbf{L}$ and $\mathbf{I}$ are respectively electron spin, electron orbital and nuclear angular momenta. These angular momenta contribute to the total atomic angular momentum and, therefore, to the total atomic dipole magnetic moment. The factors g_S, g_I and g_I are their corresponding g-factors. The field can be written in general as

$$\mathbf{B}(t) = \begin{pmatrix} B_{0x} \\ B_{0y}e^{i\delta_1} \\ B_{0z}e^{i\delta_2} \end{pmatrix} \cos(\omega t) = \hat{\mathbf{B}} \cos(\omega t) \tag{2.2}$$

It is important to keep in mind that the atom employed for this work is not a two-level system, as this is in general an approximation. Nevertheless, the coherent dynamics driven on the atom in this work acts only on a two-level subspace, allowing to consider ^{87}Rb as a two-level atom with magnetically dipole coupled states. Therefore, the Hamiltonian 2.1 is suitable.
In time-dependent perturbation theory, the total Hamiltonian is given therefore by

$$\hat{H} = \hat{H}_a + \hat{H}_B(t) \tag{2.3}$$

where $\hat{H}_a$ is the atomic Hamiltonian of the unperturbed system, with $|1\rangle$ and $|2\rangle$ as basis.
The atomic state at time t is written as

$$|\psi(t)\rangle = c_1(t)e^{-iE_1t/\hbar}\,|1\rangle + c_2(t)e^{-iE_2t/\hbar}\,|2\rangle \tag{2.4}$$

By plugging this Ansatz in Schrödinger's equation a system of coupled differential equations for coefficients $c_1(t)$ and $c_2(t)$ is obtained:

$$\dot{c_1}(t) = i\Omega_R e^{-i\phi} e^{-i\omega_2 t}\cos(\omega t) c_2(t), \tag{2.5}$$
$$\dot{c_2}(t) = i\Omega_R e^{i\phi} e^{-i\omega_1 t}\cos(\omega t) c_1(t) \tag{2.6}$$

where $\omega_1 = E_1/\hbar$, $\omega_2 = E_2/\hbar$ and Ω_R is the Rabi frequency, defined as

$$\Omega_R = \frac{\left|\langle 2|-\boldsymbol{\mu}\cdot\hat{\mathbf{B}}|1\rangle\right|}{\hbar} \tag{2.7}$$

and ϕ is the phase of the matrix element $\langle 2|-\boldsymbol{\mu}\cdot\hat{\mathbf{B}}|1\rangle = \left|\langle 2|-\boldsymbol{\mu}\cdot\hat{\mathbf{B}}|1\rangle\right| e^{i\phi}$.
By using the rotating wave approximation (namely, neglecting the terms containing $\omega_1+\omega_2$) and introducing the resonance frequency $\omega_{12} = \omega_1 - \omega_2$, the detuning $\Delta = \omega_{12} - \omega$ and the effective Rabi frequency $\Omega = \sqrt{\Omega_R^2 + \Delta^2}$, the solutions of equations (2.5)-(2.6) can be written in the following way

$$c_1(t) = \left\{c_1(0)\left[\cos\left(\frac{\Omega t}{2}\right) - \frac{i\Delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right)\right] + i\frac{\Omega_R}{\Omega}e^{-i\phi}c_2(0)\sin\left(\frac{\Omega t}{2}\right)\right\}e^{i\Delta t/2}, \tag{2.8}$$
$$c_2(t) = \left\{c_2(0)\left[\cos\left(\frac{\Omega t}{2}\right) + \frac{i\Delta}{\Omega}\sin\left(\frac{\Omega t}{2}\right)\right] + i\frac{\Omega_R}{\Omega}e^{i\phi}c_1(0)\sin\left(\frac{\Omega t}{2}\right)\right\}e^{-i\Delta t/2}. \tag{2.9}$$

The atom is assumed to be initially in the state $|1\rangle$, namely $c_1(0) = 1$, $c_2(0) = 0$. The probabilities of being in states $|1\rangle$ and $|2\rangle$ at time t are then given by $|c_1(t)|^2$ and $|c_2(t)|^2$.
Therefore, the latter results

$$|c_2(t,\Delta)|^2 = \frac{\Omega_R^2}{\Omega_R^2+\Delta^2}\sin^2\left(\sqrt{\Omega_R^2+\Delta^2}\frac{t}{2}\right). \tag{2.10}$$

The population oscillates between the two states at the effective Rabi frequency Ω, which in case of resonant driving, namely $\Delta = 0$, it boils down to the Rabi frequency Ω_R. The bigger the detuning, the faster will be the oscillations and the smaller will be the maximum population transferred to the excited state. The plots in Fig. 2.1 show the transfer probability as a function of the driving time for different detunings, and as a function of the detuning itself.
This is an important concept, since the field measurement protocol adopted with the atom, described in section 4.3.1, is based on measuring the atomic state transfer due to the interaction with MW fields.

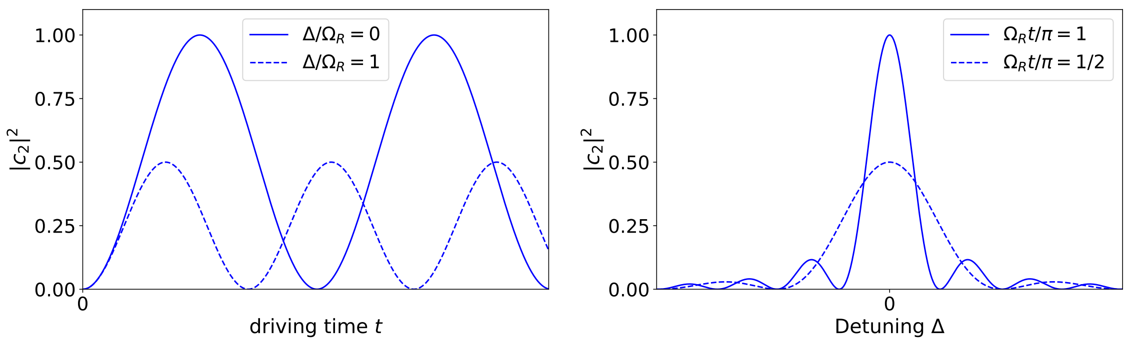


Fig. 2.1.: On the left is plotted the transition probability to the excited level $|2\rangle$ in a two level system as a function of the driving time. The solid (dashed) line represents Rabi oscillations for a resonant driving (with a detuning of $\Delta/\Omega_R = 1$).
On the right is plotted the transition probability as a function of the driving frequency, or equivalently, as a function of the detuning Δ after a fixed driving duration t. The solid and dashed line correspond to the cases $\Omega_R t/\pi = 1$ and $\Omega_R t/\pi = 1/2$. For a fixed Rabi frequency, driving the transition longer makes the peak narrower and higher.

2.2. Magnetic field polarization

The electromagnetic waves emitted by a source propagate in space and can reach regions far away from the emitter. As literature provides CITE, in such regions is possible to approximate the wavefront to a plane perpendicular to the propagation direction. A vector which helps to describe a wave is the wavevector $\mathbf{k}$, which points in the propagation direction and has magnitude equal to the wavenumber.
So, for a sinusoidal wave, the field in a point of space can be written as

$$\mathbf{B}(t) = \mathbf{B}e^{i\omega t} \tag{2.11}$$

where ω is the driving frequency and $\mathbf{B}$ is the magnetic field vector, and it describes the field polarization vector. For a completely polarized field, in two-dimensions (2D) it has the expression

$$\begin{aligned} \mathbf{B} &= \begin{pmatrix} B_x \\ B_y \end{pmatrix} = \begin{pmatrix} B_{0x}e^{i\delta_x} \\ B_{0y}e^{i\delta_y} \end{pmatrix} = e^{i\delta_x} \begin{pmatrix} B_{0x} \\ B_{0y}e^{i\delta} \end{pmatrix}, \\ \delta &= \delta_y - \delta_x. \end{aligned} \tag{2.12}$$

Therefore, it is described by four independent parameters, or three discounting an absolute phase. The vector $\mathbf{B}$ is also usually known as Jones vector, as introduced by R. C. Jones in 1941 to describe polarized light in the frame of Jones calculus [23]. Figure 2.2 shows the six common examples of normalized Jones vectors.

Another general mathematical approach to the polarized radiation in terms of its total intensity, degree of polarization, and shape parameters of the polarization ellipse was introduced by George Stokes in 1852 [24].

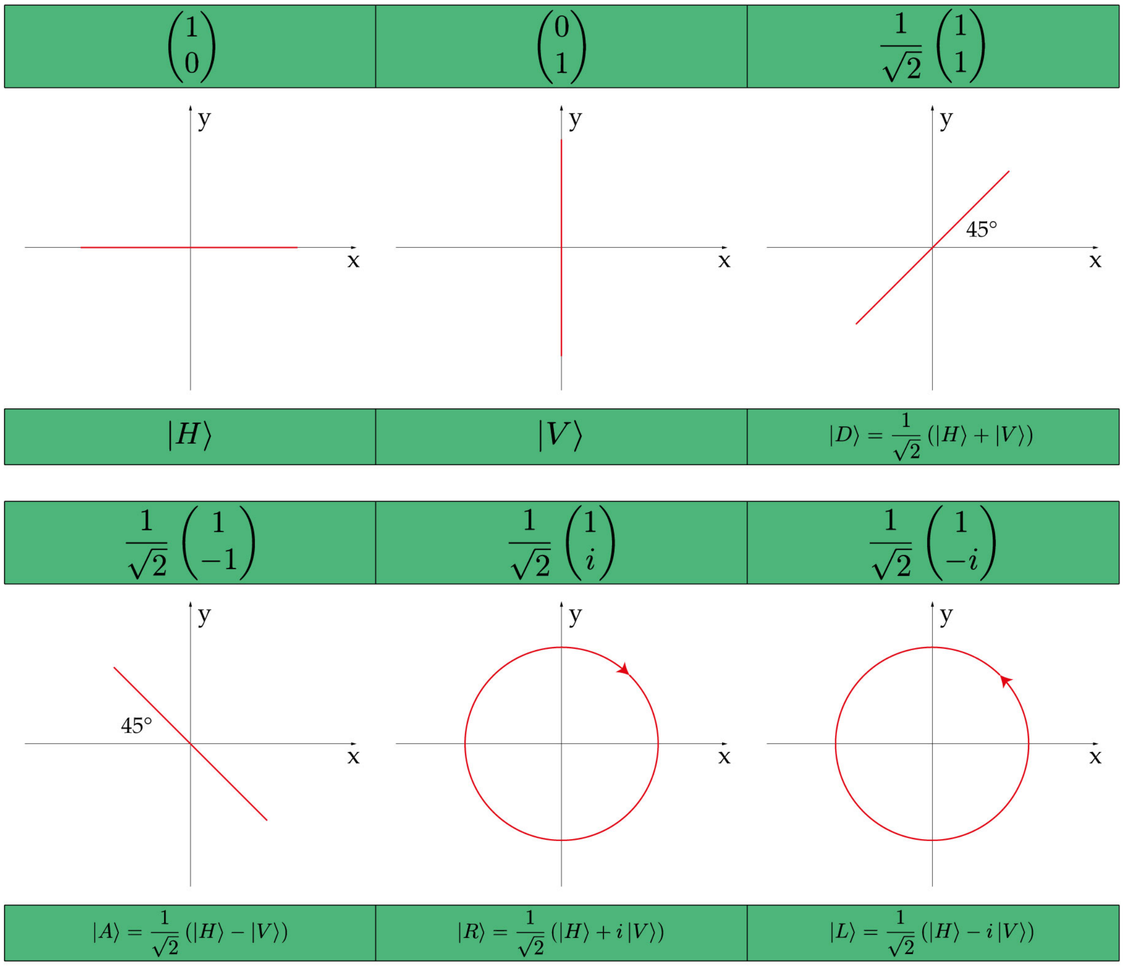


Fig. 2.2.: Six common polarization states on xy-plane. The green boxes above and below the figures contain respectively the normalized Jones vectors and the corresponding ket notation. The wavevector **k** is perpendicular to the plane and comes out of the page.

It is based on the definition of four Stokes parameters

$$S_0 = B_{0x}^2 + B_{0y}^2 \tag{2.13}$$
$$S_1 = B_{0x}^2 - B_{0y}^2 \tag{2.14}$$
$$S_2 = 2B_{0x}B_{0y}\cos\delta \tag{2.15}$$
$$S_3 = 2B_{0x}B_{0y}\sin\delta \tag{2.16}$$

which can be collected in one four-dimensional *Stokes vector*

$$\vec{S} = \begin{pmatrix} S_0 \\ S_1 \\ S_2 \\ S_3 \end{pmatrix} \tag{2.17}$$

If the magnetic field is completely polarized, its time evolution is such that the tip of the magnetic field vector describes in general an ellipse in the xy-plane. This ellipse takes the name of polarization ellipse, and any polarization state can be

(a) Representation of the field polarization in space

(b) Representation of the field polarization in the Poincaré sphere

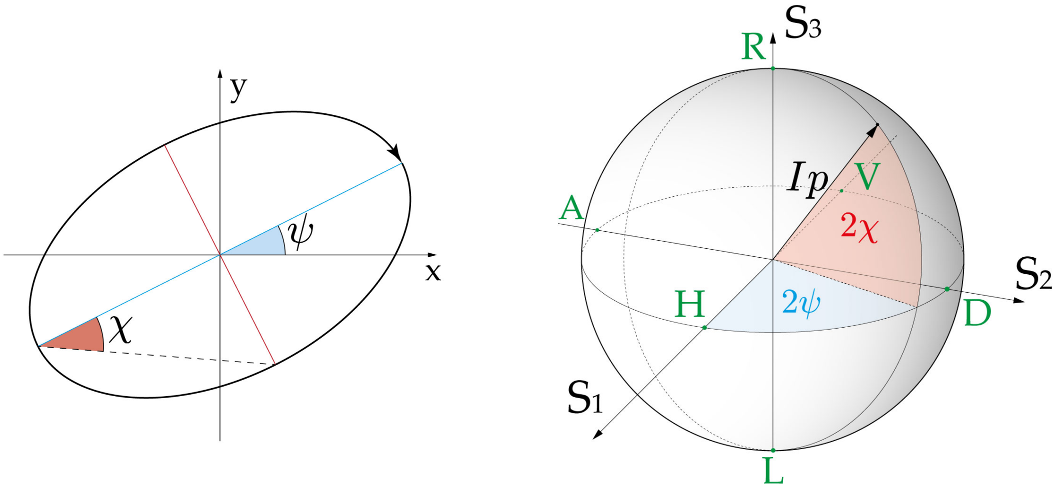


Fig. 2.3.: (a) The temporal evolution of the tip of the magnetic field vector describes an ellipse in the xy-plane. The field vector rotates in the arrowhead direction. The angles χ and ψ are the ellipse geometrical parameters, and are displayed as well in the Poincaré sphere, on the right. (b) Poincaré sphere with axes S_1, S_2 and S_3 and ray Ip, azimuthal angle 2ψ and elevation angle 2χ. Additionally, the six polarization states shown in fig. 2.2 are desplayed (green points).

described in relation to its geometrical parameters. The relationships of the Stokes parameters to intensity I and polarization ellipse parameters result

$$S_0 = I, \tag{2.18}$$
$$S_1 = Ip\cos 2\psi \cos 2\chi, \tag{2.19}$$
$$S_2 = Ip\sin 2\psi \cos 2\chi, \tag{2.20}$$
$$S_3 = Ip\sin 2\psi, \tag{2.21}$$

where p is a factor, and ψ and χ are angles. Fig. 2.3a represents an example of a polarization ellipse and shows the angles ψ and χ, while figure 2.3b provides an alternative representation of the same polarization state in the well-known *Poincaré sphere* [25]; this useful representation is justified by the identities (2.19)–(2.21), which are the equations of a sphere in spherical coordinates with ray Ip, azimuthal angle 2ψ and elevation angle 2χ.

2.2.1. Three-dimensional polarization

If two or more coherent sources emit simultaneously electromagnetic waves in the space, the fields of the different waves sum up, according to the superposition principle, resulting in interference. If each magnetic field vector lies in the same plane all the time at a given position, the interference is well described by the sum of 2D Jones vectors in that plane. The resulting polarization ellipse lies in the same plane. As soon as the magnetic fields evolve in different planes, they have to be considered as 3D-vectors and the resulting interference is well described by the sum of 3D Jones vectors. The Jones vector (2.12) is generalized in 3D as

$$
\begin{gathered}
\mathbf{B} = \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} B_{0x}e^{i\delta_x} \\ B_{0y}e^{i\delta_y} \\ B_{0z}e^{i\delta_z} \end{pmatrix} = e^{i\delta_x} \begin{pmatrix} B_{0x} \\ B_{0y}e^{i\delta_1} \\ B_{0z}e^{i\delta_2} \end{pmatrix}, \\
\delta_1 = \delta_y - \delta_x \,,\ \ \delta_2 = \delta_z - \delta_x .
\end{gathered}
\tag{2.22}
$$

Discounting the global phase as before, a number of five independent parameters, i.e. three amplitudes and two phases, describe completely the polarization state of the field. The 3D generalization of Stokes vector, according to [26], counts nine Stokes parameters in total:

$$
\begin{array}{lll}
S_0 = B_{0x}^2 + B_{0y}^2 + B_{0z}^2, & S_3 = \frac{3}{2}(B_{0x}^2 - B_{0y}^2), & S_6 = 3B_{0y}B_{0z}\cos(\delta_1 - \delta_2), \\
S_1 = 3B_{0x}B_{0y}\cos\delta_1, & S_4 = 3B_{0x}B_{0z}\cos\delta_2, & S_7 = 3B_{0y}B_{0z}\sin(\delta_1 - \delta_2), \\
S_2 = 3B_{0x}B_{0y}\sin\delta_1, & S_5 = 3B_{0x}B_{0z}\sin\delta_2, & S_8 = \frac{\sqrt{3}}{2}(B_{0x}^2 + B_{0y}^2 - 2B_{0z}^2).
\end{array}
\tag{2.23}
$$

The reason for having such a bigger number of parameters is that the Stokes vector in general spans also the space of unpolarized and partially polarized fields. However, it is opportune to specify that in this thesis only the case of full polarized fields is treated and therefore Jones vectors are sufficient for a complete polarization description.
In the 3D case, the number of independent parameters (five) exceeds sphere's dimensions (three), therefore is not possible to provide a satisfactory representation only by means of a single Poincaré sphere. nevertheless, it is always possible to associate a Poincaré sphere to an arbitrary plane in the 3D space: by considering a fixed plane, which corresponds to the choice of two out of five parameters necessary to specify completely the polarization, one can represent the polarization ellipse projected onto the plane with a Poincaré sphere. The easiest choice of a plane is trivially the plane where the ellipse lies.

2.2.2. Interference and polarization selection

The interference of the electromagnetic waves is described by the sum of the fields produced by the single sources. For a sinusoidal wave with angular frequency ω, at a given position it results

$$
\mathbf{B}_{\text{tot}}(t) = \sum_i \mathbf{B}_i e^{i(\omega t + \phi_i)}, \tag{2.24}
$$

where $\mathbf{B}_i$ and ϕ_i are respectively the Jones vector and the relative phase of the i-th source, and $\mathbf{B}_{\text{tot}}(t)$ is the total magnetic field resulting from the field's superposition. The theoretical treatment for the electric field is equivalent. Because the sum of monochromatic sinusoidal waves at the frequency ω leads to a sinusoidal wave at the same frequency, it is possible to write $\mathbf{B}_{\text{tot}}(t) = \mathbf{B}_{\text{tot}} e^{i(\omega t + \phi_{\text{tot}})}$, where $\mathbf{B}_{\text{tot}}$ and ϕ_{tot} are the Jones vector and the initial phase of the total magnetic field, respectively. Therefore, the identity (2.24) is equivalent to

$$
\mathbf{B}_{\text{tot}} e^{i\phi_{tot}} = \sum_i \mathbf{B}_i e^{i\phi_i}. \tag{2.25}
$$

Because of the experimental setup used for this thesis, presented in Chapter 3, just three distinct electromagnetic waves sources are considered. Additionally, control over amplitudes and phases of the field generated by each of them is assumed.
Thus, eq. 2.25 yields

$$\mathbf{B}_i e^{i\phi_i} = A_i \cdot \mathbf{b}_i e^{i(\psi_i + \Delta\phi_i)}, \tag{2.26}$$

where A_i and $\Delta\phi_i$ are respectively the amplitude change and phase shift manually introduced on the i-th source. Moreover, $\mathbf{b}_i$ and ψ_i are respectively the Jones vector and relative phase when $A_i = 1$ and $\Delta\phi_i = 0$, namely when there are no amplitude changes and phase shift.
Equation (2.25) can be written explicitly as

$$\begin{aligned}\mathbf{B}_{\text{tot}} e^{i\phi_{tot}} =& A_1 \cdot \mathbf{b}_1 e^{i(\psi_1 + \Delta\phi_1)} + \\ & A_2 \cdot \mathbf{b}_2 e^{i(\psi_2 + \Delta\phi_2)} + \\ & A_3 \cdot \mathbf{b}_3 e^{i(\psi_3 + \Delta\phi_3)}.\end{aligned} \tag{2.27}$$

By decomposing (2.27) in its scalar components and equating real and imaginary parts, a system of six independent equations is found. Finally, by knowing the Jones vectors $\mathbf{b}_1$, $\mathbf{b}_2$, $\mathbf{b}_3$ and the phases ψ_1, ψ_2, ψ_3 it is eventually possible to solve the system (2.27) with amplitude changes A_i and phase shifts $\Delta\phi_i$ as variables in order to obtain the desired polarization $\mathbf{B}_{\text{tot}}$.

2.3. Atomic level structure

In order to understand the effect of a field applied on an atom, it is fundamental to know its atomic level structure. In ^{87}Rb atoms, the nuclear spin $I = \frac{3}{2}$ couples to the electron spin $S = \frac{1}{2}$, giving rise to two hyperfine states $|F = 1\rangle$ and $|F = 2\rangle$ associated to the ground state $5^2S_{1/2}$, where F denotes the total angular momentum consisting of nuclear spin I, electron spin S and orbital angular momentum L.
The hyperfine energy splitting between $|F = 1\rangle$ and $|F = 2\rangle$ is 6.834 GHz, and therefore they are coupled by MW fields. Via the application of a static magnetic field, the hyperfine state manifolds can be split into $2F + 1$ magnetic Zeeman states with quantum numbers m_F where $-F \leq m_F \leq F$, as shown in figure 2.5. If the microwaves are tuned to one of the ground state transitions, and are far detuned from the other transitions, the atom is well described by a two-level system.
The excited states $5^2P_{1/2}$ and $5^2P_{3/2}$ are separated energetically by an optical transition respectively at 794 nm and at 780 nm from the ground state $5^2S_{1/2}$, and can, therefore, be neglected in the description of ground state transitions. Figure 2.4 shows the atomic-level scheme of ^{87}Rb D_2-line, which includes the manifolds $5^2S_{1/2}$ and $5^2P_{3/2}$.

2.4. Polarization-dependent coupling

As the Rabi frequency provides ad indirect measure of the magnetic field - intuitively understandable from eq. (2.7) - it is interesting to explore quantitatively the relation between the different transitions strengths and the magnetic field polarization. We are interested in the hyperfine transitions between $5S_{1/2}$ $F = 1$ and $5S_{1/2}$ $F = 2$.

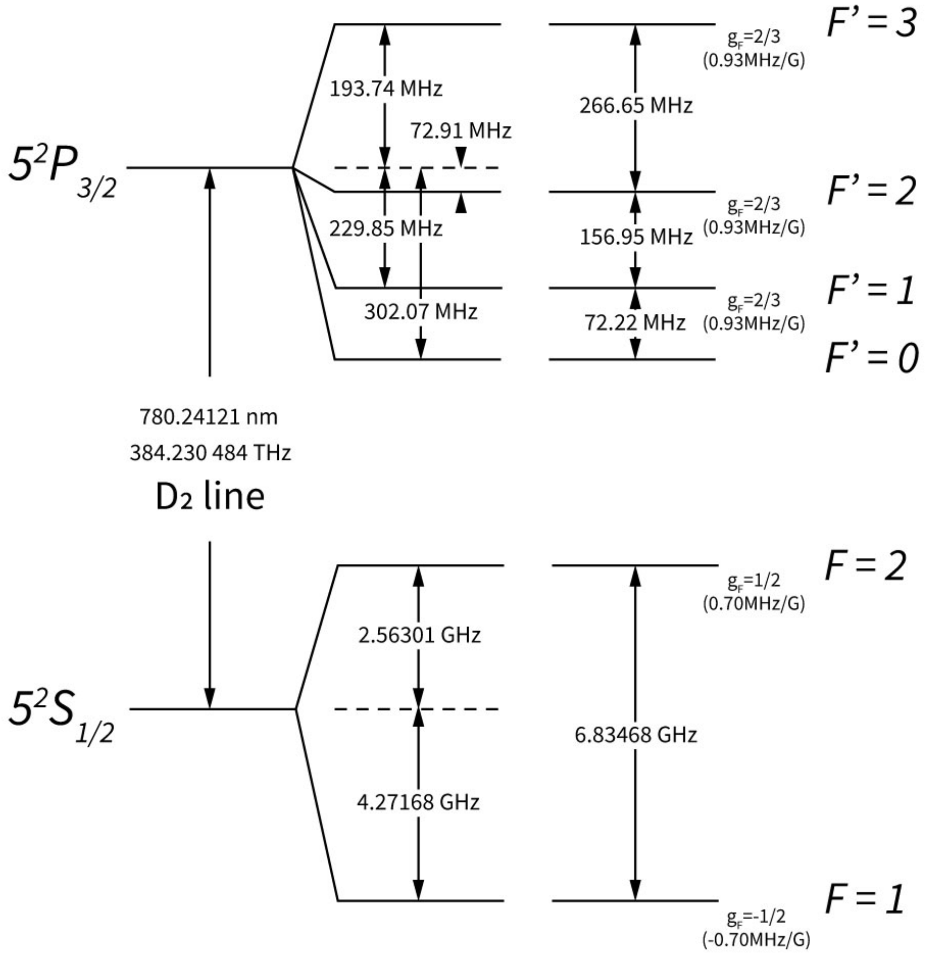


Fig. 2.4.: Atomic-level scheme of ^{87}Rb D_2-line, similar to the one reported in [21]. The hyperfine long-lived states $|F = 1\rangle$ and $|F = 2\rangle$ are separated by 6.835 GHz, whereas an optical transition at 780 nm separates the excited states in $F' = 0, 1, 2, 3$ from the two ground states. The approximate Landé g_F-factors for each level are given, which indicate the sensitivity of the energy levels to magnetic fields.

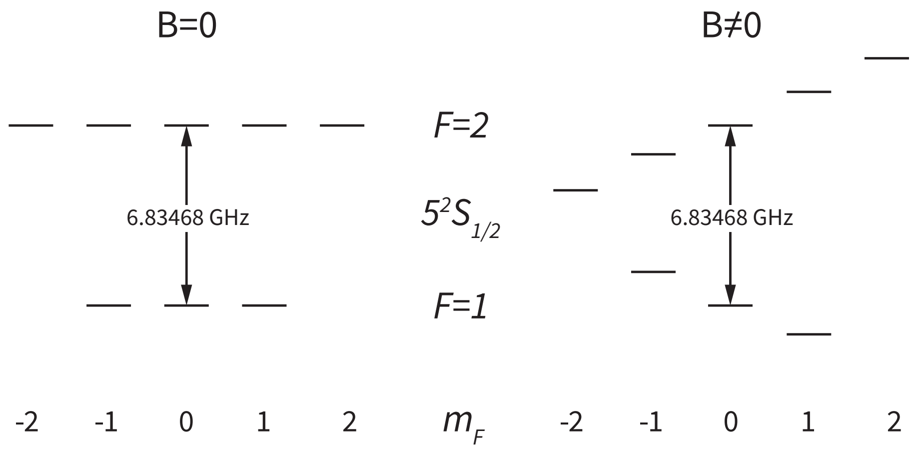


Fig. 2.5.: Zeeman splitting of ^{87}Rb hyperfine ground state manifolds induced by a static magnetic field. For reference, also the case with zero magnetic field is shown. The clock transition $m_F = 0 \leftrightarrow m_F = 0$ does not change as a result of the application of the magnetic field.

These levels show three ($m_F = -1, 0, 1$) and five ($m_F = -2, -1, 0, 1, 2$) Zeeman states respectively.
When an external static magnetic field is applied, a Zeeman splitting in the hyperfine manifolds is induced. The Hamiltonian describing the atomic interaction with the magnetic field is given by eq. (2.1). If the $5S_{1/2}$ manifold is considered, $L = 0$ and it results

$$\hat{H}_B(t) = \frac{\mu_B}{\hbar}(g_S\mathbf{S} + g_I\mathbf{I}) \cdot \mathbf{B}(t) \tag{2.28}$$

This is the magnetic dipole interaction Hamiltonian. At this point, it is important to address two questions to understand whether it is the only coupling between the manifolds $F = 1, 2$:

1. Are there magnetic multipole interactions of a higher order?
2. Are there electric multipole interactions?

In order to answer, both selection rules of electric and magnetic momenta and parity selection rules are used [27].
If with q is indicated the multiple order - i.e. $q = 1$ dipole, $q = 2$ quadrupole, $q = 3$ octupole and so on - the selection rules of electric and magnetic momenta are given by

$$|S - I| \leq q \leq S + I. \tag{2.29}$$

In ^{87}Rb atoms, the spin and orbital angular momenta are $S = \frac{1}{2}$ and $I = \frac{3}{2}$, yielding $q = 1, 2$. Therefore, according to eq. (2.29), multipole coupling of order higher than 2 between the considered manifolds are forbidden.
Parity selection rules impose the electric and magnetic multipole coupling to vanish:

- for an electric multiple, if $S + I + q =$ an odd number, (2.30a)
- for a magnetic multiple, if $S + I + q =$ an even number. (2.30b)

Applying eq.s (2.30) to ^{87}Rb, the electric (magnetic) multiple coupling vanishes if the multiple order s fulfills the equation $2+q =$ odd (even) number. Since $q = 1, 2$, only magnetic dipole and electric quadrupole coupling can exist.
In general, terms for which $q \geq 2$ are much smaller if the electric field varies little over distances comparable to the atomic radius [27]; in this case wavelengths of 4.4 cm where used, making this approximation reasonable. Therefore, the magnetic dipole operator is the dominant term, since the electric quadrupole is much smaller and, thus, negligible.
In first place, an arbitrary quantization axis is chosen, called z for convenience. Therefore, the states corresponding to this quantization axis are labeled with z. The matrix element for the π-transition between $|F=1, m_F=0\rangle_z$ and $|F=2, m_F=0\rangle_z$ is computed

$$\langle F=1, m_F=0|_z \, \hat{H}_B \, |F=2, m_F=0\rangle_z \,. \tag{2.31}$$

From the sum of angular momenta, knowing that $S=\frac{1}{2}$ and $I=\frac{3}{2}$, the Clebsch-Gordan coefficients allow to write

$$\begin{aligned}
|F=1, m_F=0\rangle_z &= \frac{1}{\sqrt{2}}\left|I=\frac{3}{2}, m_I=\frac{1}{2}, S=\frac{1}{2}, m_S=-\frac{1}{2}\right\rangle_z + \\
&\quad -\frac{1}{\sqrt{2}}\left|I=\frac{3}{2}, m_I=-\frac{1}{2}, S=\frac{1}{2}, m_S=\frac{1}{2}\right\rangle_z , \\
|F=2, m_F=0\rangle_z &= \frac{1}{\sqrt{2}}\left|I=\frac{3}{2}, m_I=\frac{1}{2}, S=\frac{1}{2}, m_S=-\frac{1}{2}\right\rangle_z + \\
&\quad +\frac{1}{\sqrt{2}}\left|I=\frac{3}{2}, m_I=-\frac{1}{2}, S=\frac{1}{2}, m_S=\frac{1}{2}\right\rangle_z .
\end{aligned}$$

For now, we cut the labels for notation convenience and stick to the representations $|F, m_F\rangle$ for Zeeman states and $|I, m_I, S, m_S\rangle$ for angular momenta.
Since the only contribution to π-transition comes from the z component of the magnetic field, we can write

$$\langle 1,0|\hat{H}_B|2,0\rangle = \frac{\mu_B}{\hbar} B_z \, \langle 1,0|(g_S\hat{S}_z + g_I\hat{I}_z)|2,0\rangle \,. \tag{2.32}$$

By using the Clebsch Gordan decomposition, the following result is derived:

$$\langle 1,0|\hat{H}_B|2,0\rangle = \frac{\mu_B}{2}(g_I - g_S)B_z. \tag{2.33}$$

2.4.1. Example: Rabi frequency for m_F=0 π-transition

From the result 2.33 it is directly possible to derive the Rabi frequency for the considered transition

$$\Omega_{|1,0\rangle\to|2,0\rangle} = \frac{\left|\langle 1,0|\hat{H}_B|2,0\rangle\right|}{\hbar} = \frac{\mu_B}{2\hbar}(g_S - g_I)B_{0z}. \tag{2.34}$$

Hence the Rabi frequency of a π-transition is directly proportional to the field amplitude along the quantization axis.

2.4.2. Coupling strength for $\sigma^{\pm}$-transitions

If $\sigma^{\pm}$ transitions are considered, B_x and B_y are the magnetic field components playing a role.
The matrix element for a σ^{+}-transition, for example $|1,0\rangle \to |2,1\rangle$ is

$$\langle 1,0|\hat{H}_B|2,1\rangle = \frac{\mu_B}{\hbar}\langle 1,0|\Big[g_S(\hat{S}_x B_x + \hat{S}_y B_y) + g_I(\hat{I}_x B_x + \hat{I}_y B_y)\Big]|2,1\rangle \tag{2.35}$$

The Clebsch-Gordan decomposition gives

$$|2,1\rangle = \frac{1}{2}\left|\frac{3}{2},\frac{3}{2},\frac{1}{2},-\frac{1}{2}\right\rangle + \sqrt{\frac{3}{4}}\left|\frac{3}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}\right\rangle \tag{2.36}$$

To compute how $\hat{S}_x, \hat{S}_y$ and $\hat{I}_x, \hat{I}_y$ act on these states, the ladder operators of the angular momentum $\hat{L}$ are recalled, namely

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y \quad \text{with} \quad \hat{L}_{\pm}|j,m\rangle = \hbar\sqrt{j(j+1)-m(m\pm 1)}\,|j,m\pm 1\rangle\,, \tag{2.37}$$

which commute with $\hat{L}^2$ and $\hat{L}_z$. Applying eq. () to $\hat{I}$ and $\hat{S}$ operators, it results

$$\hat{I}_x = \frac{1}{2}(\hat{I}_+ + \hat{I}_-),, \quad \hat{I}_y = \frac{i}{2}(\hat{I}_- - \hat{I}_+), \tag{2.38}$$
$$\hat{S}_x = \frac{1}{2}(\hat{S}_+ + \hat{S}_-),, \quad \hat{S}_y = \frac{i}{2}(\hat{S}_- - \hat{S}_+). \tag{2.39}$$

After calculations, one obtains

$$\langle 1,0|\hat{H}_B|2,1\rangle = \frac{\mu_B}{2}(g_S - g_I)\sqrt{\frac{3}{8}}\left(B_{0x} + iB_{0y}e^{i\delta_1}\right)e^{i\omega t} \tag{2.40}$$

To calculate the σ^{-}-transition, namely $|1,0\rangle \to |2,-1\rangle$, the state $|2,-1\rangle$ is written as

$$|2,-1\rangle = \frac{1}{2}\left|\frac{3}{2},-\frac{3}{2},\frac{1}{2},\frac{1}{2}\right\rangle + \sqrt{\frac{3}{4}}\left|\frac{3}{2},-\frac{1}{2},\frac{1}{2},-\frac{1}{2}\right\rangle. \tag{2.41}$$

In this case, the matrix element results

$$\langle 1,0|\hat{H}_B|2,-1\rangle = \frac{\mu_B}{2}(g_S - g_I)\sqrt{\frac{3}{8}}\left(B_{0x} - iB_{0y}e^{i\delta_1}\right)e^{i\omega t}. \tag{2.42}$$

Figure (2.6) helps visualizing the considered ground state hyperfine transitions of ^{87}Rb.

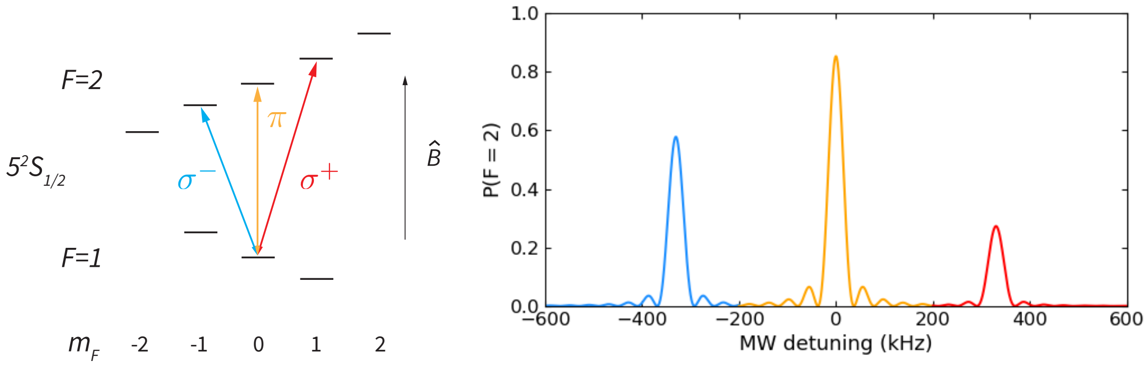


Fig. 2.6.: On the left, the hyperfine ground states splitting induced by a static magnetic field is shown. The three different couplings of the Zeeman state $|F = 1, m_F = 0\rangle$ to the $F = 2$ manifold, coinciding with microwave polarizations π, σ^+ and σ^- are depicted with different colours, which correspond to the colours of the spectra plotted on the right.
On the right, an example of transition probabilities from the state $|F = 1, m_F = 0\rangle$ to $F = 2$ for an arbitrary field polarization is plotted as function of the detuning with respect to the clock transition frequency, i.e. $|F = 1, m_F = 0\rangle \to |F = 2, m_F = 0\rangle$. These probabilities are formally expressed by eq. (2.10).

2.4.3. Ground state transition Rabi frequencies

The Rabi frequency of the π-transition $|F = 1, m_F = 0\rangle \leftrightarrow |F = 2, m_F = 0\rangle$ is given by eq. (2.34), whereas the Rabi frequencies for σ^+ and σ^- transitions, namely $|F = 1, m_F = 0\rangle \leftrightarrow |F = 2, m_F = \pm 1\rangle$, are obtained by taking the module of the computed matrix elements (2.40) and (2.42), and dividing by $\hbar$. It results

$$\pi_z : \qquad \Omega_{|1,0\rangle \to |2,0\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I) B_{0z}, \tag{2.43}$$

$$\sigma_z^+ : \qquad \Omega_{|1,0\rangle \to |2,1\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I)\sqrt{\frac{3}{8}}\sqrt{\left(B_{0x} - B_{0y}\sin\delta_1\right)^2 + B_{0y}^2\cos^2\delta_1}, \tag{2.44}$$

$$\sigma_z^- : \qquad \Omega_{|1,0\rangle \to |2,-1\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I)\sqrt{\frac{3}{8}}\sqrt{\left(B_{0x} + B_{0y}\sin\delta_1\right)^2 + B_{0y}^2\cos^2\delta_1}. \tag{2.45}$$

By choosing a second, orthogonal quantization axis - namely by applying the magnetic static field along an orthogonal direction to the previous one, called y for convenience - the effect of the phase difference δ_2 on the Rabi frequencies shows up. Repeating the calculation with the new quantization axis gives the same results as (2.33), (2.40) and (2.42), with the substitution rule

$$\begin{aligned} z &\to y, \\ y &\to x, \\ x &\to z. \end{aligned} \tag{2.46}$$

This substitution rule appears geometrically clear as soon as the reference frame is viewed from a different point of view, as shown in figure (2.7). It is also possible to

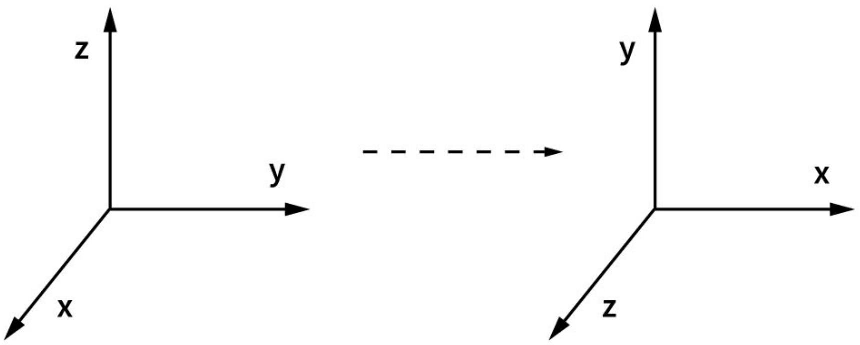


Fig. 2.7.: The shown reference frames are equivalent: they appear to be the same when looked from a different point of view, thus validating geometrically the substitution rule (2.46).

show formally that applying (2.46) the commutation relations of ladder operators of angular momentum 2.47 are fulfilled with the new quantization axis:

$$\begin{array}{c} \hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y \\ \left[\hat{L}_z, \hat{L}_\pm\right] = \pm\hbar\hat{L}_\pm \\ \left[\hat{L}_+, \hat{L}_-\right] = 2\hbar\hat{L}_z \end{array} \longrightarrow \begin{array}{c} \hat{L}_\pm = \hat{L}_z \pm i\hat{L}_x \\ \left[\hat{L}_y, \hat{L}_\pm\right] = \pm\hbar\hat{L}_\pm \\ \left[\hat{L}_+, \hat{L}_-\right] = 2\hbar\hat{L}_y \end{array} \tag{2.47}$$

Thus, labeling the states with y to indicate the y-quantization axis, the matrix elements' modules result

$$\left|\langle 1,0|_y \hat{H}_B |2,0\rangle_y\right| = \frac{\mu_B}{2}(g_S - g_I)B_{0y}, \tag{2.48}$$

$$\left|\langle 1,0|_y \hat{H}_B |2,1\rangle_y\right| = \frac{\mu_B}{2}(g_S - g_I)\sqrt{\frac{3}{8}}\left|B_{0z}e^{i\delta_2} + iB_{0x}\right|, \tag{2.49}$$

$$\left|\langle 1,0|_y \hat{H}_B |2,-1\rangle_y\right| = \frac{\mu_B}{2}(g_S - g_I)\sqrt{\frac{3}{8}}\left|B_{0z}e^{i\delta_2} - iB_{0x}\right|. \tag{2.50}$$

Finally, the Rabi frequencies for the different transitions are given by

$$\pi_y : \qquad \Omega_{|1,0\rangle\to|2,0\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I)B_{0y}, \tag{2.51}$$

$$\sigma_y^+ : \qquad \Omega_{|1,0\rangle\to|2,1\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I)\sqrt{\frac{3}{8}}\sqrt{(B_{0x} + B_{0z}\sin\delta_2)^2 + B_{0z}\cos^2\delta_2}, \tag{2.52}$$

$$\sigma_y^- : \qquad \Omega_{|1,0\rangle\to|2,-1\rangle} = \frac{\mu_B}{2\hbar}(g_S - g_I)\sqrt{\frac{3}{8}}\sqrt{(B_{0x} - B_{0z}\sin\delta_2)^2 + B_{0z}\cos^2\delta_2}. \tag{2.53}$$

2.5. AC Zeeman shift

In general, when an atom is probed with quasi-resonant light, the atomic energy levels are shifted due to the electric dipole coupling to the oscillating perturbation-field. The electric field component of a nearly-resonant laser induces a so-called AC Stark shift on the transition frequency, which is well known in the literature [28]. The ^{87}Rb ground states are magnetic dipole coupled, excluding the presence of an

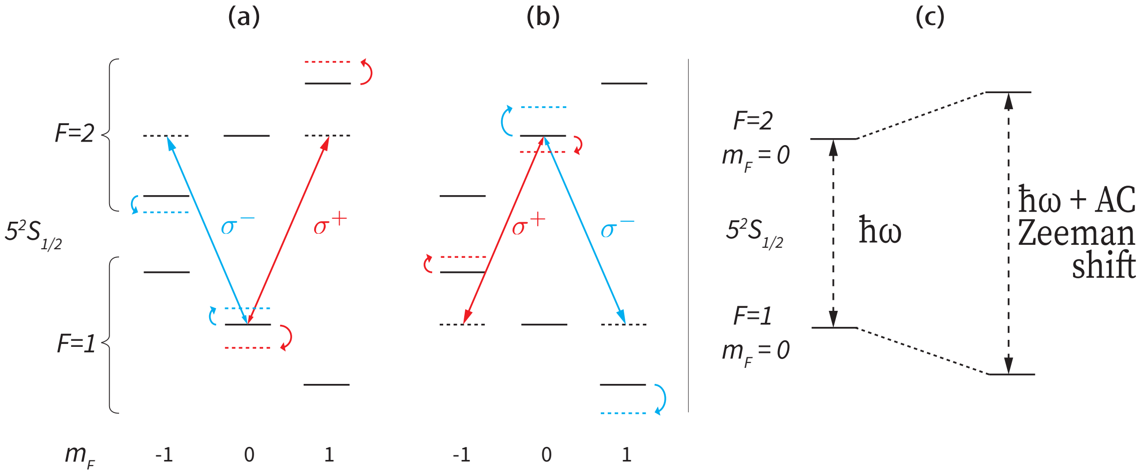


Fig. 2.8.: Sketch of the energy shifts of ^{87}Rb ground states due to the presence of a MW magnetic field tuned to the transition $5^2S_{1/2}$ $F = 1, m_F = 0 \longleftrightarrow F = 2, m_F = 0$. The Zeeman levels with $m_F = \pm 2$ are not considered because they do not play a role. (a) Here, only the couplings between the states $F = 1, m_F = 0$ and $F = 0, m_F = \pm 1$ are considered. The σ^+ and σ^- components of the MW field induce an AC Zeeman shift on the state $F = 1, m_F = 0$ of opposite signs, because of the opposite detunings. (b) Here, only the couplings between the states $F = 2, m_F = 0$ and $F = 1, m_F = \pm 1$ are considered. These couplings have in general a different strength from those of fig. 2.8a. A shift on the state $F = 2, m_F = 0$ is induced. (c) The sum of the shifts of the states $5^2S_{1/2}$ $F = 1, 2$ $m_F = 0$ results in the clock transition frequency shift.

AC Stark effect. Nevertheless, by analogy with the electric field, we make the educated guess that the interaction between MW magnetic field and atomic ground state magnetic dipole induces an analogous shift in the ground state hyperfine transition frequencies. Since it is due to the magnetic field, this shift takes the name of AC Zeeman shift.
In particular, if the clock transition $5^2S_{1/2}$ $|F = 1, m_F = 0\rangle \longleftrightarrow |F = 2, m_F = 0\rangle$ is considered, the microwaves will have to be tuned on a shifted resonance frequency, due to the coupling of $|F = 1, m_F = 0\rangle$ and $|F = 2, m_F = 0\rangle$ to the other Zeeman states $|F = 2, m_F = \pm 1\rangle$ and $|F = 1, m_F = \pm 1\rangle$, respectively. As it is possible to see in figure 2.8, these states are coupled only by σ^+- and σ^--polarized MW fields. Therefore, by suppressing σ^+ and σ^- polarizations, the frequency shift can be extinguished. The theory to quantify this effect is discussed in the following. If a two-level atom is considered, with levels identified by $|e\rangle$ and $|g\rangle$, the AC Zeeman shift δE_g of the energy level $|g\rangle$ can be obtained for instance by a density matrix approach [29]. This yields

$$\delta E_g = -\frac{1}{4}\hbar\,\mathrm{Re}\left\{\frac{|\Omega_{eg}|^2}{\omega_{eg} - \omega - i\gamma/2} + \frac{|\Omega_{eg}|^2}{\omega_{eg} + \omega + i\gamma/2}\right\}, \tag{2.54}$$

where the quantities are defined in section 2.1 and the transition decoherence rate γ is introduced.
For an atom with a multilevel structure of excited levels, the shift becomes

$$\delta E_g = -\frac{1}{4}\hbar\sum_e \mathrm{Re}\left\{\frac{|\Omega_{eg}|^2}{\omega_{eg} - \omega - i\gamma/2} + \frac{|\Omega_{eg}|^2}{\omega_{eg} + \omega + i\gamma/2}\right\}, \tag{2.55}$$

where the sum runs over the excited states, labeled with e.
Therefore, the frequency shift $\delta\omega_{|1,0\rangle}$ of the ground state $|1,0\rangle$ is given by

$$\delta\omega_{|1,0\rangle} = \frac{\delta E_{|1,0\rangle}}{\hbar} = -\frac{1}{4}\sum_{m_F=-1,0,+1}\mathrm{Re}\left[\frac{|\Omega_{|2,m_F\rangle-|1,0\rangle}|^2}{\omega_{|2,m_F\rangle-|1,0\rangle}-\omega-i\gamma_{|2,m_F\rangle-|1,0\rangle}/2}+\frac{|\Omega_{|2,m_F\rangle-|1,0\rangle}|^2}{\omega_{|2,m_F\rangle-|1,0\rangle}+\omega+i\gamma_{|2,m_F\rangle-|1,0\rangle}/2}\right]. \tag{2.56}$$

Here, $\omega_{|2,m_F\rangle-|1,0\rangle} = \omega_{2,m_F} - \omega_{1,0}$ and the decoherence rate $\gamma_{|2,m_F\rangle-|1,0\rangle}$ depends on the transition considered. Analogously, the frequency shift $\delta\omega_{|2,0\rangle}$ state $|2,0\rangle$ is given by

$$\delta\omega_{|2,0\rangle} = \frac{\delta E_{|2,0\rangle}}{\hbar} = -\frac{1}{4}\sum_{m_F=-1,0,+1}\mathrm{Re}\left[\frac{|\Omega_{|1,m_F\rangle-|2,0\rangle}|^2}{\omega_{|1,m_F\rangle-|2,0\rangle}-\omega-i\gamma_{|1,m_F\rangle-|2,0\rangle}/2}+\frac{|\Omega_{|1,m_F\rangle-|2,0\rangle}|^2}{\omega_{|1,m_F\rangle-|2,0\rangle}+\omega+i\gamma_{|1,m_F\rangle-|2,0\rangle}/2}\right] \tag{2.57}$$

Thus, the AC Zeeman frequency shift of the clock transition $\delta\omega_{|2,0\rangle-|1,0\rangle}$ results

$$\delta\omega_{|2,0\rangle-|1,0\rangle} = \delta\omega_{|2,0\rangle} - \delta\omega_{|1,0\rangle}. \tag{2.58}$$

This expression has been computed and plotted in fig. 2.9a as a function of MW frequency. Fig. 2.9b shows how the simultaneous suppression of σ^+ and σ^- polarization leads the AC Zeeman frequency shift to vanish. However, an experimental confirmation is pending.

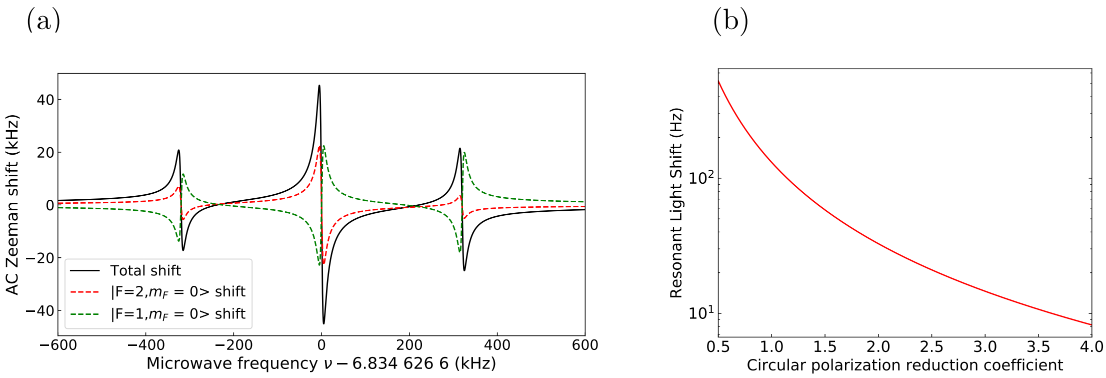


Fig. 2.9.: (a) AC Zeeman frequency shift of the clock transition $0-0$. For this computation, the results represented in the upper right plot of Fig. 4.9 are used. The Rabi frequencies are $\Omega_\pi = 2\pi\times\ 12\,\mathrm{kHz}$, $\Omega_{\sigma^-} = 2\pi\times\ 9\,\mathrm{kHz}$, $\Omega_{\sigma^+} = 2\pi\times\ 11\,\mathrm{kHz}$, whereas the decoherence rates are assumed to be $\gamma \approx 2\pi\times\ 10\,\mathrm{kHz}$ for all transitions. Resonances appear on the central frequency and on the spectator transitions, involving σ^+ and σ^- microwave polarization. (b)AC Zeeman shift with microwaves resonant on the clock transition as function of the circular polarization reduction coefficient, defined as the factor by which Ω_{σ^+} and Ω_{σ^-} are simultaneously reduced. Thus is possible to suppress the resonant AC Zeeman Shift by proper manipulation of microwaves polarization.

3. Microwave Antennas Test Setup

The central goal of this thesis is to develop a method to select the MW field polarization. In order to achieve this goal, we can distinguish some general main steps:

- being able to emit the microwaves and to measure the polarization;
- building an apparatus that allows to tune the MW fields interference in order to obtain the desired polarization;
- finally proving to have a reasonable match between predicted and measured polarization.

Premised that the final aim is to drive atomic transitions in single ^{87}Rb atoms trapped in a cavity, these steps are independent of the specific experimental realization considered. In this work, the accomplishment of these steps is explored both in a test apparatus and, independently, with atoms in a cavity setup - see sec. 4. Depending on the specific physical system, a different "polarization probe" is required. The separate test apparatus is composed by loop antennas as MW emitters and dipole antennas as receivers. It was built for two main reasons:

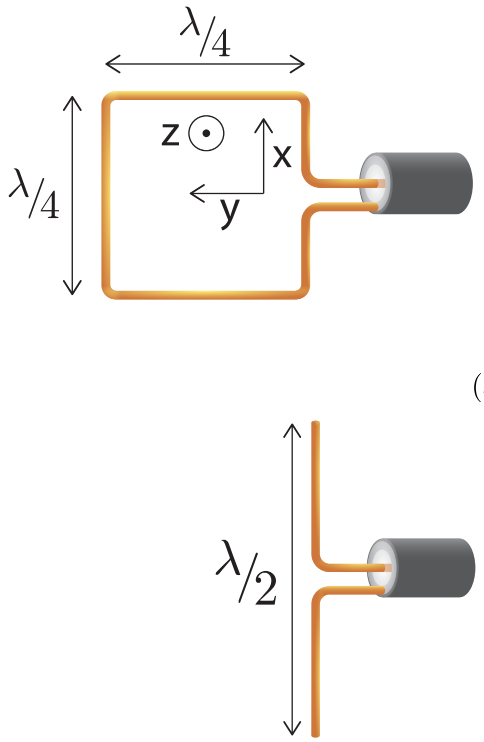
1. test the method to select a defined MW field polarization outside the chamber before applying it directly to the cavity setup;
2. characterize the devices used to amplify and control MW fields and check their reliability.

This chapter is dedicated to the test setup, where constitutive elements, calibration techniques and faced issues - and solutions - are detailed. The strategy employed to predict and achieve a defined polarization is discussed and the obtained results are presented - together with a representation of a Poincaré sphere applied to polarization in 3D.

3.1. Constitutive elements

A full scheme of the test setup is shown in fig. 3.6, where its constitutive elements are labeled and described in detail in this Section.
As emitters, it was made use of three MW loop antennas (3.6a) made out of copper, schematically represented in fig. 3.1a. The antennas are built and positioned as similarly as possible to those present in the vacuum chamber. They are square self-resonant single-loops, namely, they consist of a single turn of conductor and their circumference equals the MW wavelength used. This implies that the antennas transfer function has a resonance at the considered frequency. Additionally, the antennas are placed perpendicularly to each other.

In this case the MW frequency is $\nu = 6.834$ GHz, thus the length is approximately 4.4 cm. The choice of three antennas perpendicular to each other is aimed to have a sufficient number of degrees of freedom to reach all polarizations.
If a loop is small in comparison to its operating wavelength - typically circumference below 30% of a wavelength - it is equivalent to a magnetic dipole along the axis normal to the plane of the loop [30], with the feature of having the direction of maximum transmission and reception within the plane of the wires. Large loops show the opposite feature, having maximum signal broadside to the wires since the unsuitability of the magnetic dipole approximation leads to a more complicated radiation pattern. An approximate solution of very good accuracy for the square-loop antenna can be found in [30] and projections onto antenna's principal planes are plotted in fig. 3.2. Additionally has to be considered that imperfections in antennas fabrication distort this pattern; moreover, albeit in the case of precise radiation pattern's knowledge, such a piece of information is not sufficient to predict precisely the field in some point of the space, due to the presence of environment reflections, scattering and absorption, which can become significant effects.

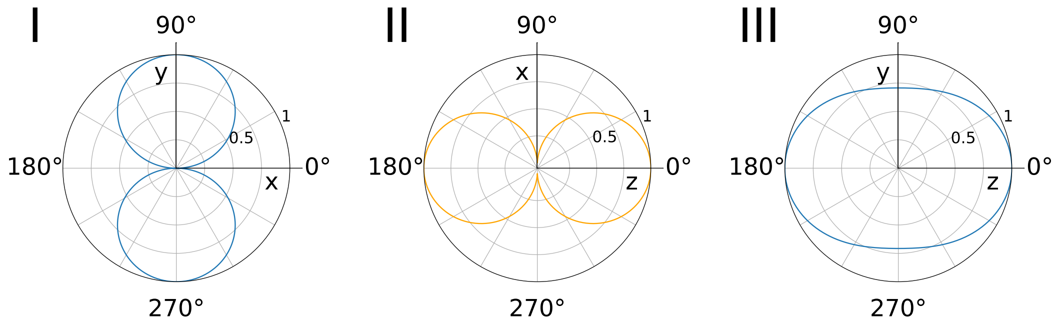


(a)
(b)

Fig. 3.1.: (a) Square self-resonant loop, with circumference of 4.4 cm. (b) Half-wave dipole antenna, with total length 2.2 cm.

Devices commonly used as MW emitters are MW horns - antennas that consist of a waveguide shaped like a horn to direct waves into a beam. The reason why loop antennas are used in the vacuum chamber instead of horn antennas, is that horns for 6.8 GHz would require much more space in the chamber. In the past, a horn antenna was used successfully in the same laboratory outside the chamber to drive MW transitions in ^{87}Rb atoms. Nevertheless, fast transitions could not be achieved because high MW power would have caused a overheating of the chamber and of the equipment hit by the radiation. As MW receivers, three orthogonally aligned dipole

(a)



(I) xy-plane (II) zx-plane (III) zy-plane

(b)

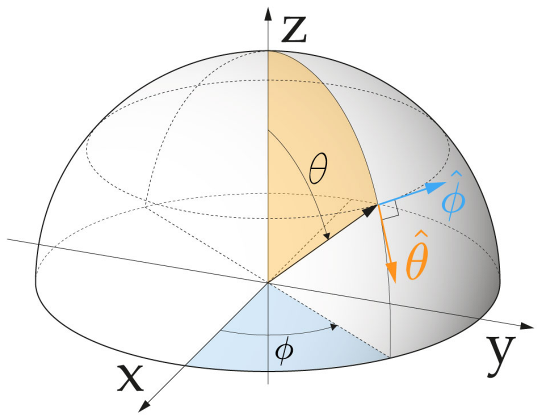


Fig. 3.2.: (a) Square loop antenna radiation pattern projected onto its principal planes, spanned by the coordinate axes given in fig. 3.1a. The radiation pattern is completely described by electric field's components along $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ - see fig. b - respectively E_θ and E_ϕ.
(I) In blue the xy-plane normalized pattern plot of E_ϕ, whereas $E_\theta = 0$.
(II) In orange the zx-plane normalized pattern plot of E_θ, whereas $E_\phi = 0$.
(III) The zy-plane normalized pattern plot of E_ϕ, whereas $E_\theta = 0$.
(b) Radiated fields observed far from any antenna decay with distance as a spherical wave ($\propto \frac{1}{r}$), whereas the variation with observation angles (θ, ϕ) depends on the size and construction details of the antenna [30]. Thus it is sufficient to study field's angles dependencies. A convenient way is to investigate field's components along θ and ϕ, namely E_θ and E_ϕ.

antennas (3.6b) were used. A dipole antenna commonly consists of two metal rods of the same length fed at the center. Half-wave dipoles were fabricated, depicted in fig. 3.1b, since the fundamental resonance of a thin linear conductor occurs when the wavelength is twice the wire's length. The signal detected by a dipole antenna is proportional to the electric field component along the antenna's direction. Therefore, a set of three orthogonal dipole antennas allows a measurement of relative field amplitudes and phases along these directions. This accomplishes the complete measurement of the field's polarization, as further discussed in Section 3.2. Fig.

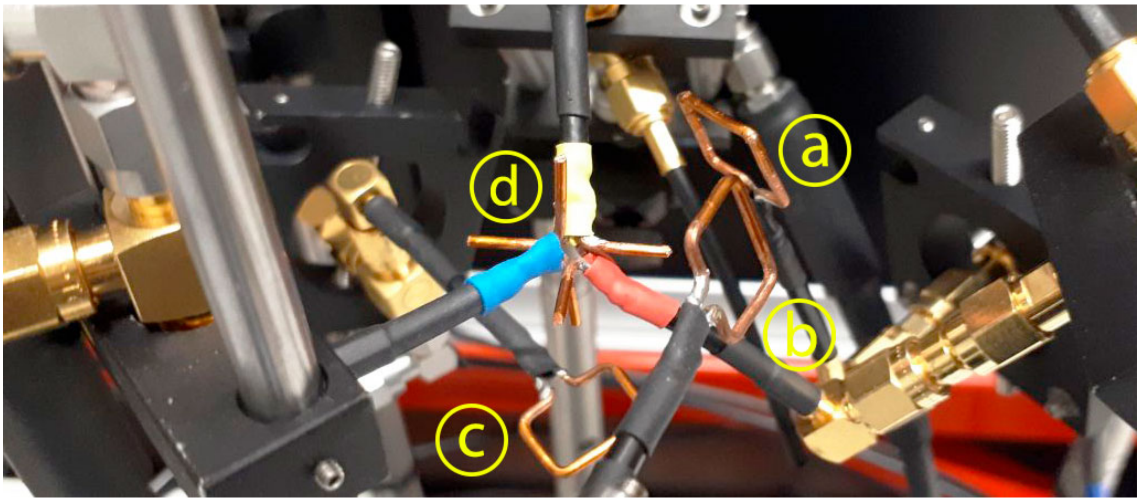


Fig. 3.3.: The three square loop antennas (a), (b) and (c), work as MW emitters. A set of three orthogonal dipole antennas (d) is employed as MW receiver. When hit by the MW fields emitted by the loops, dipole antennas produce a signal proportional to the electric field component along dipole's direction.

3.3 shows a photograph of all emitters and receivers in the built test setup. Dipole antennas were placed as symmetrically as possible, in order to reduce asymmetrical radiation distortion due to their proximity. Additionally, it can be noticed that even though loop antennas were not placed perfectly perpendicularly, good results were achieved with this apparatus - as described in Section 3.3.

As discussed in Section 2.2.2, the ability of controlling relative phases among the three antennas and amplitudes of their feeding signals is crucial to create interference in order to achieve the desired polarization. This is made possible by the use of variable phase shifters (3.6c) and attenuators (3.6d) operating at 6.834 GHz. Specifically, two voltage control phase shifters *RF-LAMBDA RVPT0408GBC* and three Continuous Variable Attenuators *QUALWAVE QCA10-2-18-40* - the latter controlled by mechanical rotation of a micrometer screw gauge - were employed. Fig. 3.4 shows measured attenuation and phase shift as functions of screw length or voltage at the working input powers. Additionally, these curves depend also on the input power.
It is important to notice that phase shifters introduce, in addition to the phase shift, an attenuation depending on the voltage. Analogously, attenuators introduce, in addition to the attenuation, a phase shift depending on the micrometer screw position. Therefore, if an attenuator and a phase shifter are placed along the same line, these features contribute to the total attenuation and to the total phase shift in the line. Due to the absence of variable attenuators when the test apparatus measurements were performed, fixed attenuators were used instead. The attenuation introduced either by an attenuator or a phase shifter was obtained by measuring the losses in a spectrum analyzer centered at 6.834 GHz - see fig. 3.5a.
The phase shifts measurement method is depicted in fig. 3.5b and relies on the detection of phase differences between two signals at 10 MHz - enabled by down mixing two MW signals differing by 10 MHz.
Signal's amplification was provided through fixed +35 dB amplifiers (3.6)e, *MICROWAVE AMPS AM53-6.5-7-37-37* and water cooling technique was imple-

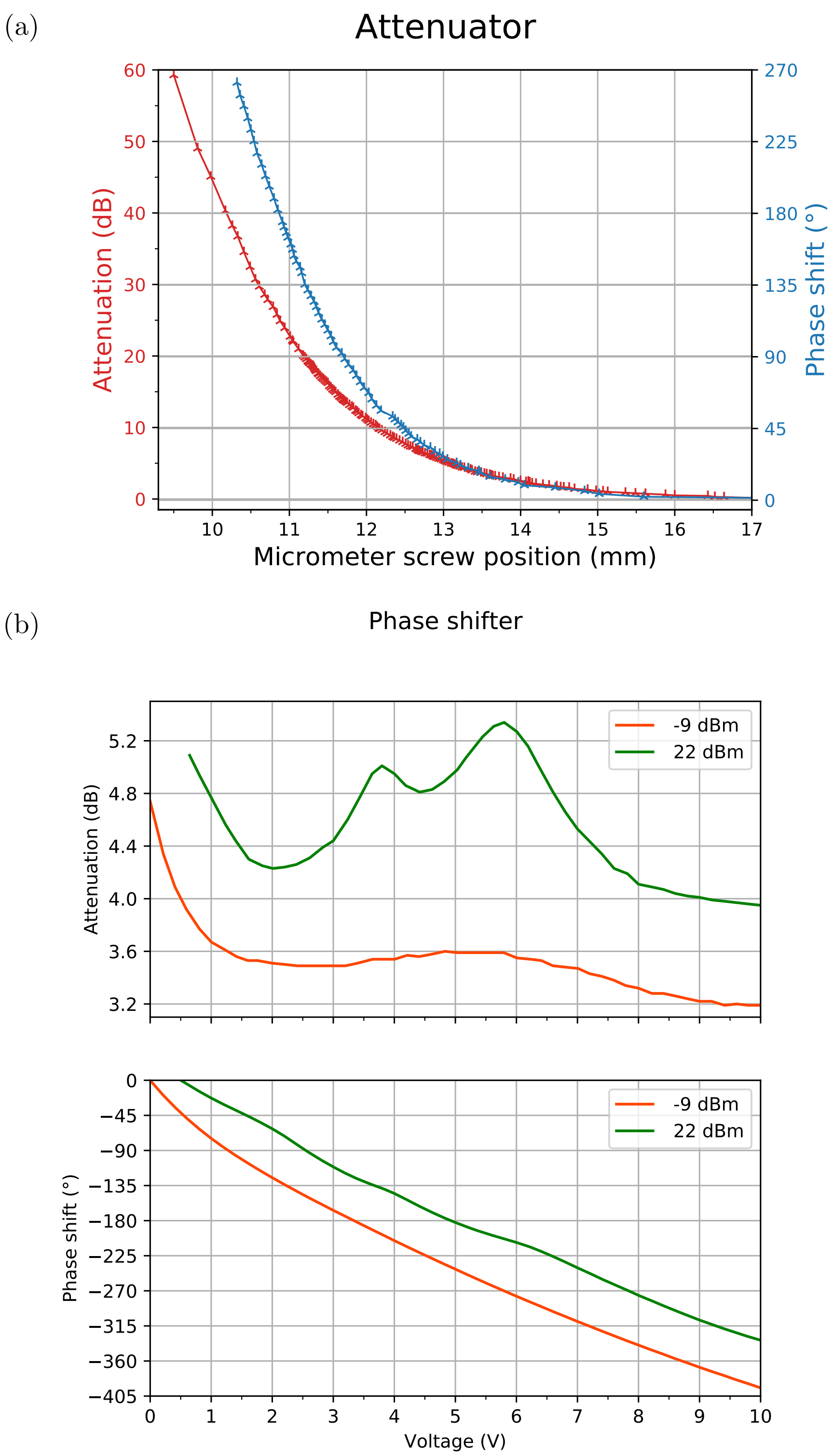


Fig. 3.4.: (a) Typical attenuation and phase shift of a continuously variable attenuator as function of the micrometer screw position at 6.834 GHz and 23 dBm in input. Error bars are given by instruments precision and are smaller than the linewidth. (b) variable control phase shifter's attenuation and phase shift as function of the voltage at 6.834 GHz at different input powers. Error bars are given by instruments precision and are smaller than the linewidth.

(a)

(b)

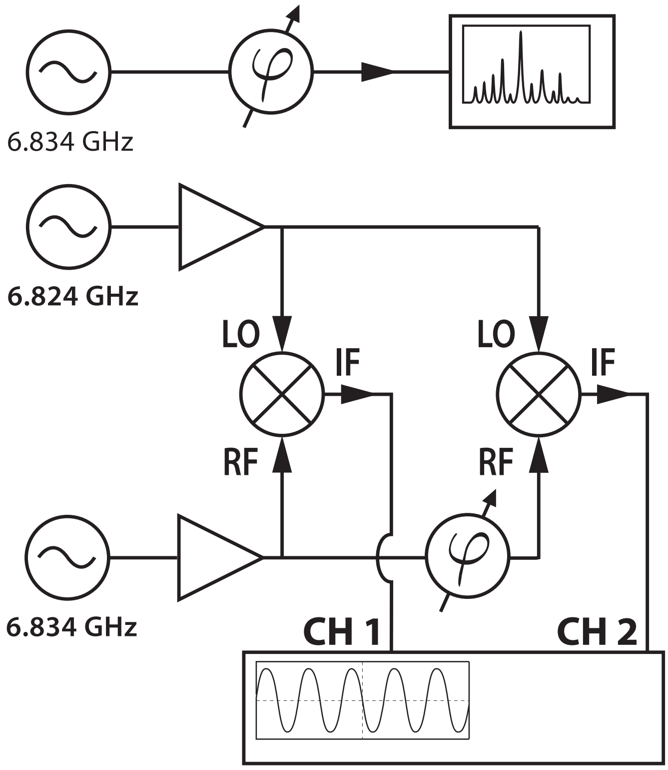


Fig. 3.5.: (a) Method for measuring the attenuation introduced by a device - represented with the variable phase shifter symbol in this scheme: the power of the MW signal is measured at device input and output via a spectrum analyzer centered at 6.834 GHz. The attenuation is given by their difference in dB.
(b) Method for measuring the phase shift introduced by a device: two signals coming from two MW sources differing by 10 MHz are split in two, mixed and read on an oscilloscope. The bandwidth of the oscilloscope only allows to measure the signal at the difference of the original frequencies. The signal at oscilloscope CH1 serves as a reference, while phase shifts introduced by the device translate directly into phase shifts in CH2. For both methods, amplification and attenuation in the lines are adjusted in order to have the desired input power.

mented to prevent their overheating. To this end, the amplifiers were originally equipped with fans, which were introducing mechanical noise at about 50 Hz, corresponding to fans' rotation's frequency.

As it will become clearer in Section 3.2, it is preferred to avoid as much as possible the cross-talk between the output of the signal splitters, due to undesired back-reflections occurring to the signals traveling toward the antennas. To this end, MW circulators (3.6f) were used - where back-traveling signals are redirected to a third port and absorbed by an appropriate termination (*WEVERCOMM WVQ-6.8G-0.8GBW02*).
Finally, in order to read the 6.834 GHz MW signals detected by the dipole antennas, they were mixed by frequency mixers (3.6g) *MINI CIRCUITS ZMX-8GLH* with a 6.824 GHz sinusoidal signal, resulting in 10 MHz signals which are measured by the oscilloscope.

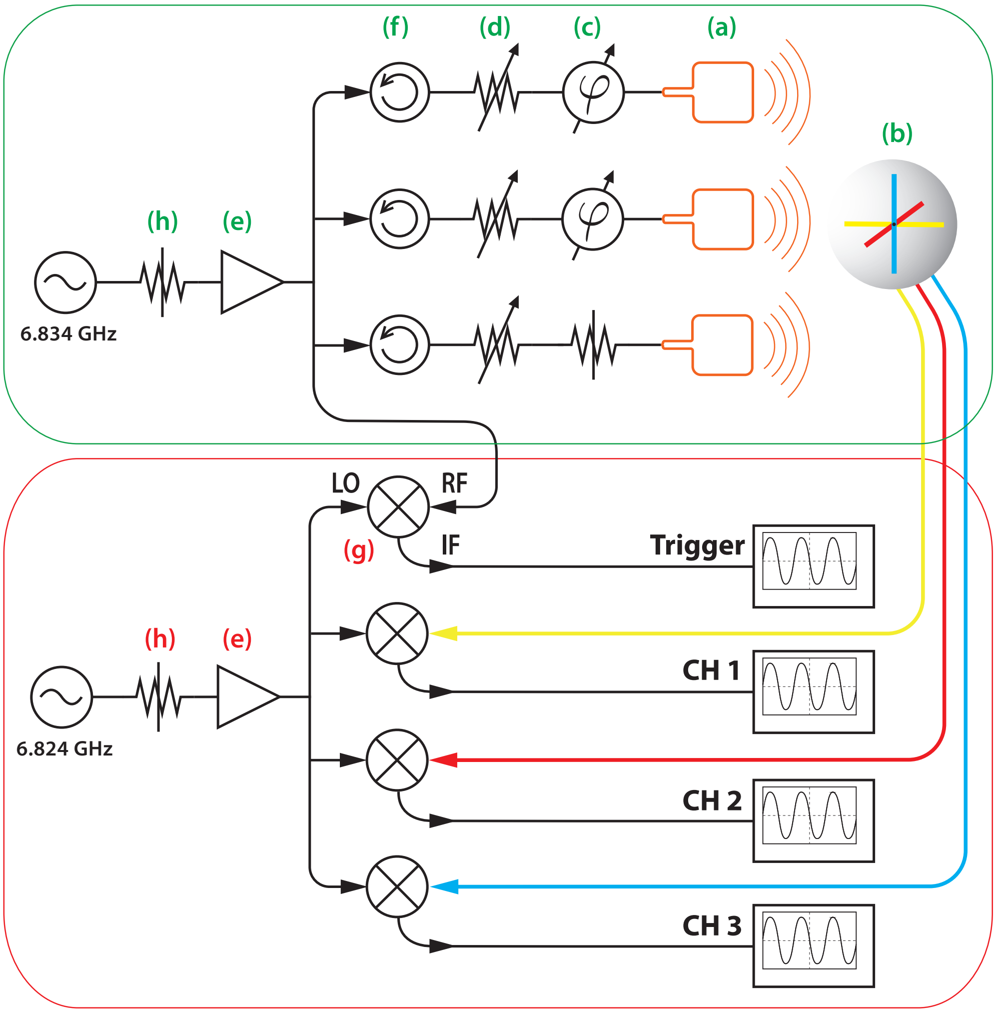


Fig. 3.6.: Full MW antennas test setup scheme with all constitutive elements. The green box encloses apparatus section containing the elements generating, controlling and detecting the MW fields, whereas the red box surrounds the section hosting devices used for the measurement. The signals are finally measured by an oscilloscope. The separated oscilloscope symbols refer all to the same device.
a) Self-resonant square loop antennas as MW field emitters at 6.834 GHz;
b) Three orthogonal half-wave dipole antennas as MW field receivers: MW cables - depicted in red, blue and yellow, corresponding to dipole antennas axes - transmit the signals from the antennas to the mixer (g) LO ports;
c) Voltage controlled phase shifters;
d) Variable attenuators;
e-e) Amplifiers;
f) Circulators to limit the cross-talk among the three feeding lines;
g) Mixers: 6.834 GHz in RF ports and 6.824 GHz in LO ports resulting in 10 MHz output signals, measured by the oscilloscope;
h-h) Fixed attenuators to reduce the input power;

3.2. Polarization control method

According to equation (2.27), the achievement of the desired Jones vector relies on the knowledge of the fields generated by the single sources and their phase relation and the ability of controlling amplitudes and relative phases among sources. Since the test setup measures electric field components along x, y and z direction - respectively in red, yellow and blue in figures 3.3 and 3.6 - it is appropriate to rewrite for the electric field

$$
\begin{aligned}
\mathbf{E}_{\text{tot}} = & A_1 \cdot \mathbf{e}_1 e^{i(\psi_1+\Delta\phi_1)} + \\
& A_2 \cdot \mathbf{e}_2 e^{i(\psi_2+\Delta\phi_2)} + \\
& A_3 \cdot \mathbf{e}_3 e^{i(\psi_3+\Delta\phi_3)},
\end{aligned}
\tag{3.1}
$$

where now $\mathbf{e}_i$ is the Jones vector of the electric field produced by the i-th loop antenna and ψ_i are the phases of the Jones vectors. These quantities are determined by the orientation and the spatial position of the antennas, and can not be changed because the emitters arrangement is fixed. Additionally, A_i and $\Delta\phi_i$ are respectively the amplitude variation and phase shift that can be manually introduced by means of attenuators and phase shifters.
At this point, it is good to point out that even though the electric field and not the magnetic field is measured, which is the relevant one for the atomic transitions considered in this thesis, electromagnetic waves carry both fields, which obey to the superposition principle and thus the discussion in Section 2.2 applies to both.
Now the goal is to measure the Jones vectors and phases of the MW fields produced by the emitters, and to use this information to tune phase shifts and attenuations.

Step 1: measurement of Jones vectors

The measurement of $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ is carried out in the following way. After having decided on a chosen configuration, i.e. fixing antennas positions, the setup environment, attenuators and phase shifters, single fields $\mathbf{e}_i$ are measured by feeding the antennas one by one, by disconnecting manually the other two lines.
The 3D Jones vector (2.22) is rewritten for the electric field as

$$
\mathbf{E} = \begin{pmatrix} E_{0x} \\ E_{0y} e^{i\delta} \\ E_{0z} e^{i\delta'} \end{pmatrix}
\tag{3.2}
$$

and is completely characterized by three amplitudes E_{0x}, E_{0y} and E_{0z} and two relative phases δ and δ'. The output of the detected 6.834 GHz MW signal mixed with a 6.824 GHz sinusoidal signal in measured on an oscilloscope, thus measuring the amplitudes and the relative phases - see fig. 3.6. The result of a measurement is shown in fig. 3.73.7a. Here, only a single antenna is emitting a MW field. The same measurement is taken for the two remaining antennas.

Step 2: measurement of Jones vector phases

In order to derive the phases ψ_1, ψ_2 and ψ_3, the three antennas are driven simultaneously - maintaining the same configuration as before - and the field produced by single fields superposition is measured. Therefore

(a) (b)

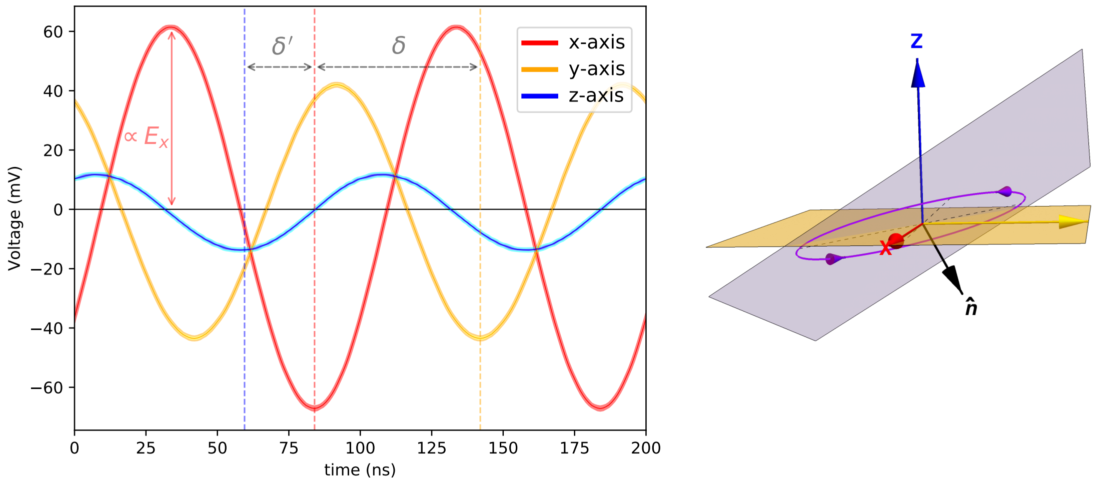


Fig. 3.7.: (a) Typical data taken with an oscilloscope. The three 10 MHz signals coming from the mixers are read by the oscilloscope, retrieved from the dipole antennas placed along three orthogonal labeled axes, shown in fig. 3.3 and 3.6. According to eq. (3.2), δ and δ' are directly derived by signal phase differences, while the amplitudes are proportional to the electric field components. The standard deviation error bars are thinner than the linewidth.
(b) Corresponding 3D polarization ellipse. The reference frame is given by the coloured axes, labeled as in fig. 3.3 and 3.6; the polarization ellipse is shown in violet and the dotted lines are the major and minor axes of the ellipse. At every time the electric field lies always in the polarization ellipse plane (violet) and rotates in the direction given by the arrowheads along the ellipse. To give the feeling of 3-dimensionality, both vector $\hat{\mathbf{n}}$ normal to the polarization plane, in black, and xy-plane in transparent yellow are shown.

$$\mathbf{E}_{\text{tot, ref}} = \mathbf{e}_1 e^{i\psi_1} + \mathbf{e}_2 e^{i\psi_2} + \mathbf{e}_3 e^{i\psi_3} \tag{3.3}$$

gives the total field $\mathbf{E}_{\text{tot, ref}}$ when all antennas are driven in the reference configuration. By measuring it on the oscilloscope, it is possible to derive directly the relative phases from the equation (3.3).

Step 3: tuning of phase shifts and attenuations

Finally, it is possible to solve the system of six scalar equations (3.1) with A_i and $\Delta\phi_i$ as variables, after having chosen the desired polarization $\mathbf{E}_{\text{tot}}$. Lastly, phase shifters and attenuators are tuned to achieve the wanted A_i and $\Delta\phi_i$.
To tune them correctly, it is important to recall that both introduce both phase shifts and attenuation , characterized in fig. 3.4. Moreover, it is crucial to adjust the input power in order not to exceed the mixer 1dB compression point, defined as the compression power in RF port at which mixer's loss increases by 1 dB - introducing a non-linear behavior in the measurement procedure and therefore making it no

longer reliable.
An other error source concerning the variation of phase shifters and attenuators control parameters - i.e. the voltage for the phase shifters and the screw's position for the attenuators - has to be pointed out: due to an imperfect input impedance matching of these devices, the Voltage Standing Wave Ratio (VSWR) changes with their control parameters and so does the back-reflected power at their inputs. While these additional return losses are included in attenuation measurements 3.4, some amount of power turns out in the other feeding lines, which is difficult to take into account and to compensate for. This problem can nevertheless be bypassed by placing circulators (3.6f) at the feeding lines starting points, thereby preventing the undesired cross-talk, sketched in fig. 3.8.

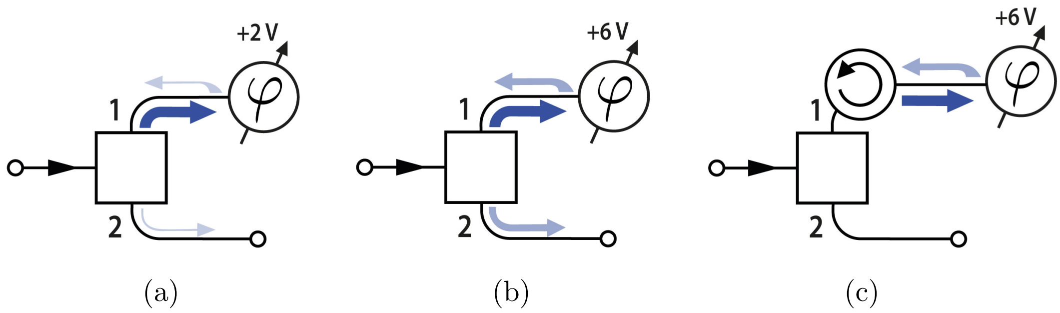


Fig. 3.8.: Imperfect devices VSWR causing cross-talk between lines at splitter output ports. (a) The outgoing signal from port 1 - thick blue arrow - is partially back-reflected at phase shifter input, enters port 1 and travels attenuated in port 2 - thin light blue arrows. (b) When phase shifter's control voltage is changed the VSWR varies aswell, and so does the back-reflected power - thicker light blue arrow. The same happens to attenuators when screw's length is changed. (c) The reflected power is deviated by the circulator and absorbed by a proper termination.

3.3. Test apparatus results

In order to characterize the performance of this setup as polarization controller, six polarization states were measured and their Jones vectors were compared with the predicted ones. In particular, exploiting the analogy with the 2D scenario - treated in Section 2.2 - the six cardinal points on the Poincaré sphere were measured - displayed in fig. 2.2. It is crucial to recall that a single Poincaré sphere can correspond only to a single plane in space, where the polarization ellipse lies; in the 3D case, the polarization ellipse could in principle lie on an arbitrary plane, allowing no longer to use the Poincaré sphere representation.
Nevertheless, it is possible to fix an arbitrary plane in the space, where the six polarization states 2.2 can exist. At this point, the 3D polarization can be represented in this plane by projecting the polarization ellipse onto the plane. Therefore, the resulting polarization state can exist in the Poincaré sphere corresponding to that plane. The chosen projective Poincaré sphere is shown in fig. 3.9 together with the predicted and resulting measured polarization states. In order to quantify the match

between theoretical expectations and measured states, the overlap $\mathcal{O}$ of predicted and prepared Jones vectors was computed as

$$\mathcal{O} = \left|\mathbf{B}_{\mathrm{theor}} \cdot \mathbf{B}_{\mathrm{mes}}\right|^2 . \tag{3.4}$$

The overlap of the six polarization states are

H: 99.7(2) % V: 97.7(6) %
D: 99.7(1) % A: 99.0(2) %
R: 98.2(8) % L: 98.9(6) %

yielding an arithmetic mean of 98.9(5) %. When the measurements were performed, it was possible to apply attenuations only in discrete steps, namely via fixed attenuators. As a result of this, the predicted states move away from the cardinal points - red points in fig. 3.9.

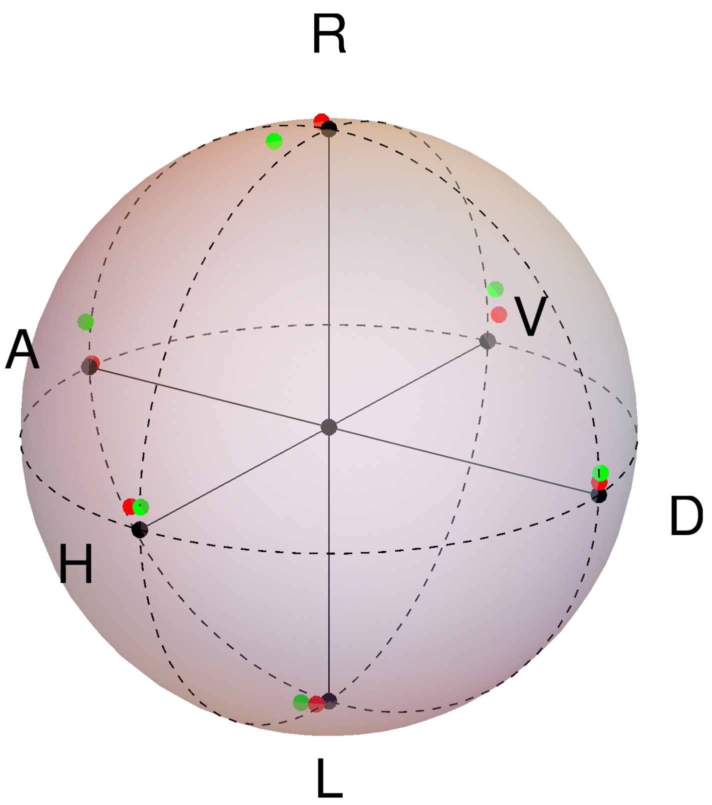


Fig. 3.9.: Predicted (red) and prepared (green) states are represented on a Poincaré sphere. It can be clearly noticed that the predicted states do not coincide perfectly with the cardinal points: this is due to employment of fixed attenuators instead of continuous attenuators. The impossibility to use the exact required attenuation makes theoretical points move away from the cardinal points.

4. Microwave Setup in the Vacuum Chamber

This Chapter focuses on the setup and the techniques to trap, prepare and perform measurements on atoms in the vacuum chamber. Here, single neutral ^{87}Rb atoms are trapped in a double cavity setup: MW field polarization can not be probed by receiver antennas. Thus the strategy to infer and measure selected polarizations involves measurement methods performed directly on the atom, typical of quantum optics field: in this case, MW Rabi oscillation measurements and MW spectroscopy are the fundamental ingredients.

4.1. Crossed fiber cavities

The core of the setup is composed of two fiber cavities, whose modes cross perpendicularly. Fig. 4.1 shows the crossed fiber cavities, post-colored for visualization purposes. They consist of two single-mode fibers with a cladding diameter of 125 µm, a multi-mode fiber with a cladding diameter of 125 µm and a multi-mode fiber with a cladding diameter of 200 µm. The reasons for the employment of these fibers are explained in [31], such as cavities production, characterization and features. By convention, they are called short and long cavity because of their different length

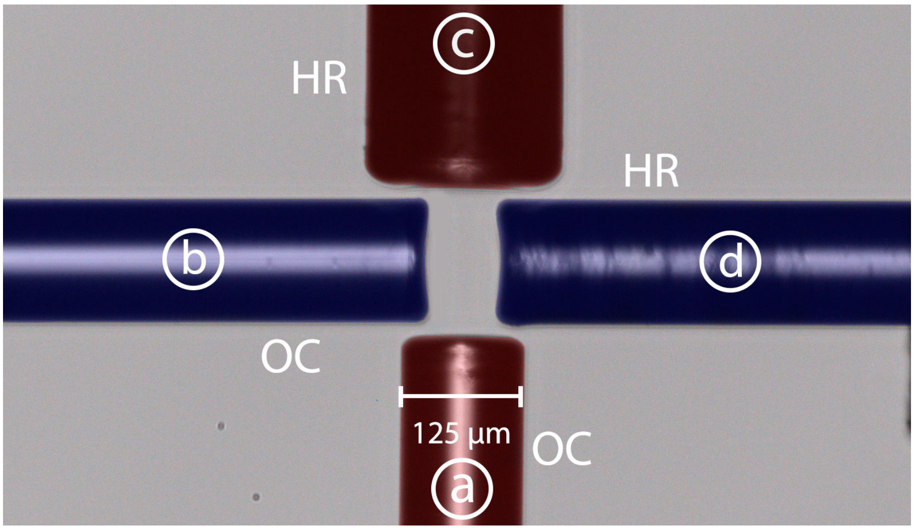


Fig. 4.1.: Colored photograph of the crossed fiber cavities. The short one (blue) has a length of 80 µm, whereas the long one (red) has a length of 160 µm. Fibers (a) and (b) are single mode and couple the light into the cavities. Fibers (b) and (d) are multi mode and have high reflective mirrors. The abbreviations *OC* and *HR* refer to the outcoupling mirror and the highly-reflective mirror, respectively.

- respectively 80 µm and 160 µm approximately. The use of crossed optical fiber cavities coupled to a single atom in quantum information experiments opens new possibilities in the realization of nodes in quantum networks, for example allowing the heralded and passive storage of photonic qubits by means of a single atom [32].

4.1.1. Atoms in crossed fiber cavities

For this project, ^{87}Rb-atoms have been used. These are initially provided from a dispenser, which heats up through current and releases atoms in the vacuum chamber, and trapped by a magnetic-optical trap (MOT), which is roughly 10 mm above the horizontally aligned cavity plane. During the loading phase of the MOT, a cooling laser and a repumping laser are used simultaneously. The former is 30 MHz red-detuned to the transition $5S_{1/2}$, $F = 2 \longleftrightarrow 5P_{3/2}$, $F' = 3$ and shines a power of 10 mW, while the latter is 20 MHz red-detuned to the transition $5S_{1/2}$, $F = 1 \longleftrightarrow 5P_{3/2}$, $F' = 2$ with a power of 250 µW.
After a 2 s long MOT-loading phase, the corresponding laser lights and magnetic field are shut down, the cold atoms cloud falls due to the gravitational force and expands because of its finite temperature.

The atoms are subsequently trapped in the center of the cavities via one vertical dipole trap and two intracavity-dipole traps. Shortly after releasing the MOT, a standing-wave dipole trap is shined in vertically through the chamber, passing through the MOT and the crossing-point of the cavities modes. This laser light is 4.2 nm red-detuned to the D_1-line ($5S_{1/2} \longleftrightarrow 5P_{1/2}$, transition at 795 nm) and it serves as guiding beam for the cold cloud to the cavities crossing-point, in order to increase the atom loading probability into the cavities, and at the same time as standing-wave red-detuned dipole trap.
The intracavity traps are coupled in through the fibers themselves and are both blue-detuned to the transition $5S_{1/2} \longleftrightarrow 5P_{3/2}$. For the long (short) cavity, a wavelength of 774.6 nm (776.4 nm) has been used, with a typical power of 400 nW (200 nW). The advantage of having blue-detuned traps is that the atoms are trapped in standing-wave nodes, thus reducing the rate with which they scatter the light of trap lasers. Powers are adjusted to a $\sim$ 1 mK deep trap depth.

Once the atoms reach the cavity modes crossing-point, additional cooling beams are shined, in order to be able to dissipate atoms energy, and, consequently, to trap them. Part of the fluorescence light produced during the cooling process is collected by the cavities and detected by single photons counting modules (SPCMs). In this way is possible to detect the presence of atoms coupled to the cavities.
The cooling beam is positioned along a direction that is oblique to the three trap beams, with a power of roughly 200 nW tuned to the transition $5^2S_{1/2}$, $F = 2 \longleftrightarrow 5^2P_{3/2}$, $F' = 3$. Moreover, a repumper beam is shined along the same direction with typically 30 nW on the transition $5^2S_{1/2}$, $F = 1 \longleftrightarrow 5^2P_{3/2}$, $F' = 2$. This additional laser light ensures that the population decaying from $F' = 3$ to $F' = 2$ is repumped in the cooling cycle, thus making it more efficient.

The experiment starts by preparing the atom in the Zeeman state $5^2S_{1/2}$ $F = 1$, $m_F = 0$ via optical Zeeman pumping. The latter uses cooling laser light on the transitions $F = 2 \longleftrightarrow F' = 2$ and $F = 2 \longleftrightarrow F' = 3$, in order to depopulate

the manifold $F = 2$. Additionally π-polarized light resonant with the transition $F = 1 \longleftrightarrow F' = 1$ is shined along the vertical axis, with the aim to pump out the states $F = 1$, $m_F = \pm 1$ with higher rate, whereas $F = 1$, $m_F = 0$ is electrical-dipole-prohibited and stays dark, thus accumulates population.
If the preparation in a Zeeman state is not required - or, in some cases, is not possible - hyperfine pumping is performed. As opposed to Zeeman pumping, this does not use π-polarized light resonant with $F = 1 \longleftrightarrow F' = 1$, causing the population to accumulate in the full manifold $F = 1$. Once the state preparation is accomplished, the MW pulses on the transition $F = 1 \longleftrightarrow F = 2$ can be applied and the resulting population in $F = 2$ can be measured via state detection.

A method to perform state detection is based on the possibility that a hyperfine ground state can be coupled to an excited state via a closed transition, thereby scattering multipole photons, whereas the other hyperfine-ground states are far detuned and therefore appear dark [33]. The transition $5S_{1/2}$, $F = 2 \longleftrightarrow 5P_{3/2}$, $F' = 3$ is closed, making therefore the manifold $F = 2$ the easiest choice for the bright state. Moreover, the short cavity tuned to the mentioned transition enhances the fluorescence emission in the cavity mode via the Purcell effect [34].

In order to ensure that a single atom is trapped, sequences of $g^{(2)}(\tau)$ correlation measurements are performed intermediately during the cooling sequence and before the state initialization. The internal atom state is at first prepared in the state $F = 2$ via optical pumping on the transitions $F = 1 \longleftrightarrow F' = 1$ and $F = 1 \longleftrightarrow F' = 2$.

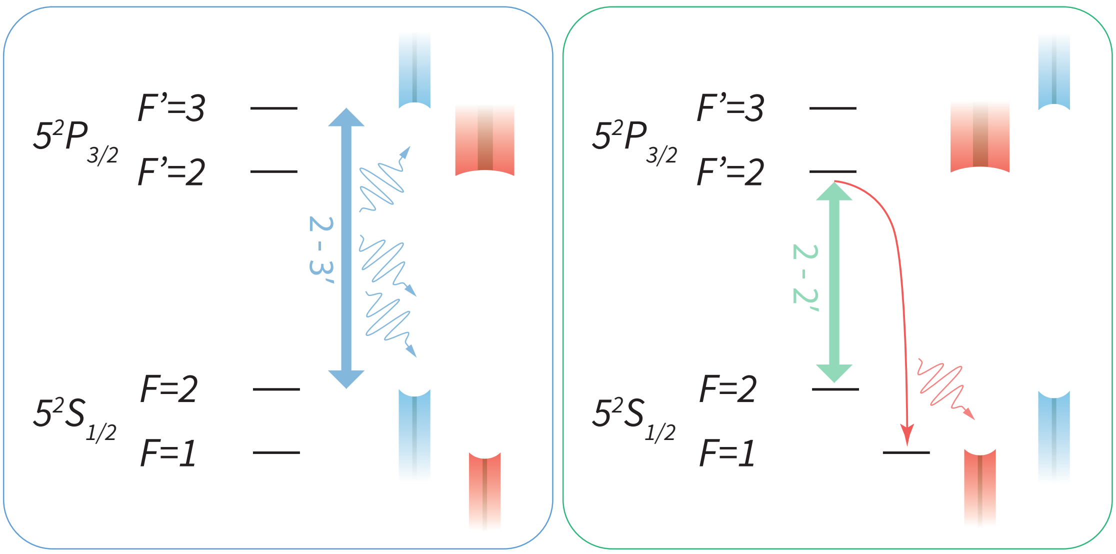


Fig. 4.2.: Shown in the left box is the hyperfine state detection method. Laser light is shined on the closed transition $F = 2 \longleftrightarrow F' = 3$ and the fluorescence photons are dominantly collected by the short cavity, tuned nearly in resonance on the same transition - drawn, as for Fig. 4.1, in blue. The long cavity, drawn in red, is far detuned and therefor does not play a role for the state detection.
Shown in the right box is the "single" photon emission method useful to the $g^{(2)}$ measurement. The atom is pumped back and forth on the transition $F = 2 \longleftrightarrow F' = 2$ while the long cavity is tuned on $F = 1 \longleftrightarrow F' = 2$, thus enhancing the emission of a photon into the long cavity mode during the pumping process.

Subsequently, a short laser pulse on $F = 2 \longleftrightarrow F' = 2$ is shined in, typically of 2 µs duration, while the long cavity is tuned on $F = 1 \longleftrightarrow F' = 2$, enhancing the spontaneous emission of a single photon in the cavity mode. At this point the atom is in $F = 1$, which is by now dark and can no longer be excited to $F' = 2$, suppressing the emission of further photons in the cavity. The emitted photons are measured using a Hanbury-Brown Twiss interferometer in order to evaluate $g^{(2)}(0)$, from which the number of atoms n is inferred via $g^{(2)}(0) = 1 - 1/n$.

It is interesting to notice that, as opposed to the state detection, the single-photon emission is based on the use of a non-closed transition, combined with the enhanced decay on a dark state due to the cavity vacuum field, whereas for the state detection the short cavity is resonant to the cycling transition in order to collect fluorescence light. Fig. 4.2 summarizes state detection and "single" photon production procedures on the atom.

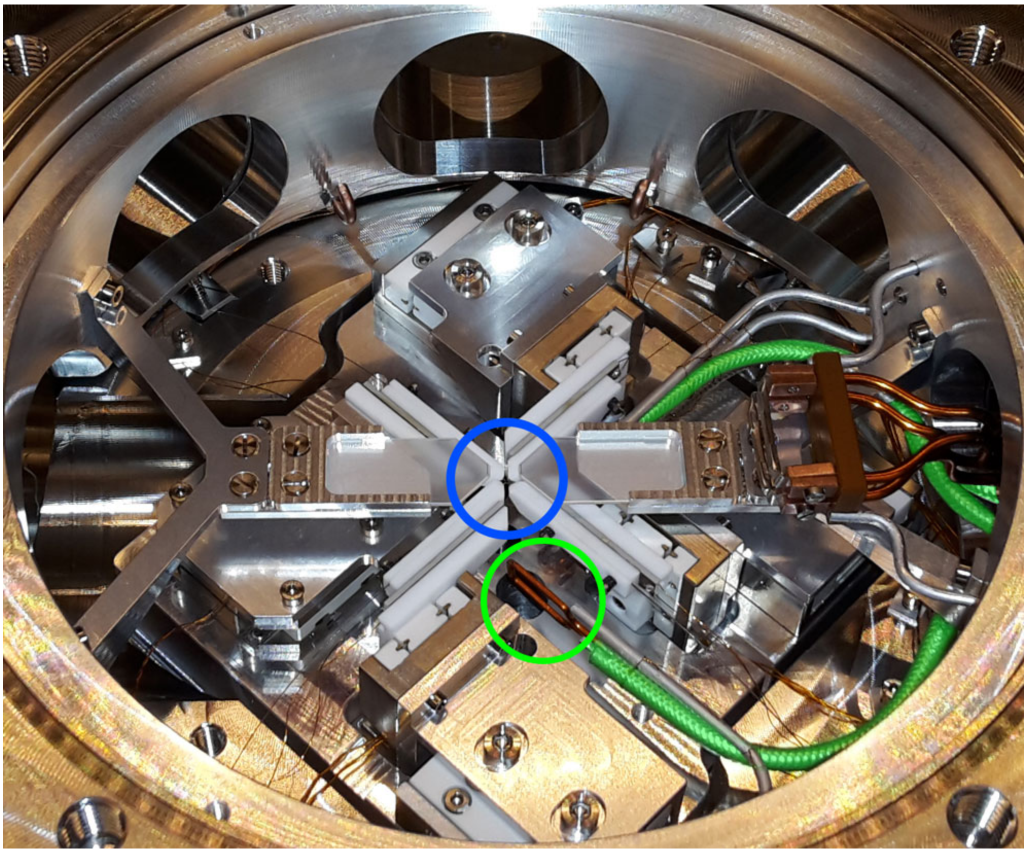


Fig. 4.3.: Photograph of the pieces integrated in the vacuum chamber. The crossed cavities are placed in the center of the chamber - in the blue circle, much smaller than the circle. The coaxial cables feeding the antennas are colored in green - only partially visible - whereas an antenna is encircled in green.

4.2. Microwave setup

Three MW square loop antennas of circumference 4.4 cm are installed in the vacuum chamber in order to drive the hyperfine ground state transitions of ^{87}Rb at 6.8 GHz. They are built out of Kapton-isolated copper wire and are connected to semi-rigid coaxial cables (*ALLECTRA 380-SMA18G-MM-1000* suitably cut, based on *312-PTFE-Coax-SR*), with specified maximum attenuation of 1.5 dB/m at 6.8 GHz.
The antennas are placed along three orthogonal directions, such that the control over phases and amplitudes of MW fields allows in principle to drive polarization-resolved hyperfine ground state transitions. They are placed as close as possible to the atom (smallest distance of $\sim$ 6 mm) and, at the same time, far enough so that they don't touch the cavity holders and don't obstacle any laser beam. Figure 4.3 shows a view of the chamber where is possible to see a MW antenna.
The MW signal, provided by a MW synthesizer (*R&S SMA100B - SIGNAL GENERATOR*), is split in the three antennas channels with an arrangement similar to the one described in section 3.1 and shown in the green box of Fig. 3.6. The main difference is the presence of amplifiers and two circulators in each line, as sketched in Fig. 4.4.

In order to maximize the transmitted power from the feeding lines to the antennas, an impedance matching circuit might be needed. In this particular case, a stub-tuning technique has been used to match antennas input impedance with the 50 Ω lines output impedance. Given the Cartesian form of antennas input impedance $Z = R + jX$, where R is the resistance and X is the reactance, the method consists in nulling the reactance and making the resistance R =50 Ω by placing a stub at the input and modifying its length and insertion point [35], as depicted in Fig. 4.5a.
By looking at the reflected power at the involved frequency, the stub has been cut until a reflection minimum was reached, whereas the insertion point was left unvaried. The impossibility to modify iteratively the length of the stub and its insertion point together represents the main drawback for this method, being the reached minimum very likely just a local minimum. Nevertheless, an improvement in back-reflections suppression has been done, as reported in Fig. 4.5b. Even though the power transmitted to the antenna has been increased more than an order of magnitude (+12 dB), measurements on the atomic transitions showed that the power increase was not appreciable on the atom.

To check the power stability, two out of three antennas have been used to pick-up MW pulses of ∼11 µs emitted by the third antenna. Both antenna outputs are fed into a mixer, which outputs rectangular pulses whose area is dependent on the MW fields phase relation - assumed to be constant - and picked-up power. Figure 4.6 reports the histogrammed pulse areas, measured directly by an oscilloscope. The presented Gaussian distribution shows a relative error σ/μ =0.7%.
In order to drive successfully MW transitions by using all three antennas together, it is fundamental that the MW pulses are synchronized. This has been tested by down-mixing them over different paths, including different electrical elements as shows the sketch in Fig. 4.7a, and calculating the cross-correlation function on the pulses measured by the oscilloscope. The time cross-correlation function is defined as

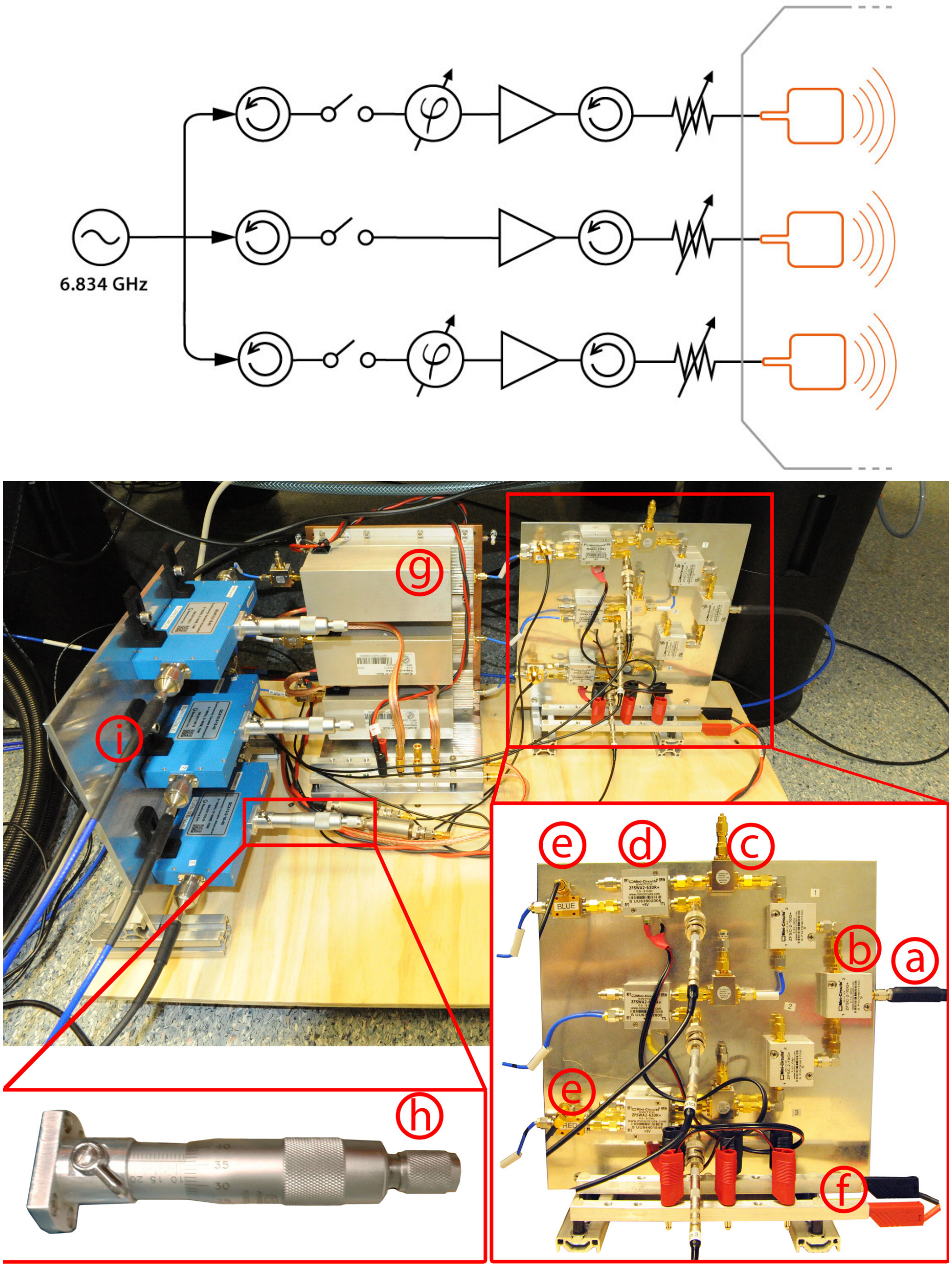


Fig. 4.4.: On the top, sketch of the MW feeding system. The three antennas are installed in the chamber, depicted by the gray enclosure on the right. Additionally to the circulators at the beginning of each line, whose role is explained in Fig. 3.8, further ones are appended to each amplifier, in order to avoid the damage of the amplifiers by back-reflected signals. Moreover, every line contains a fast transistor-transistor logic (TTL) controlled switch.
On the bottom, photograph of the MW feeding system. The MW signal in input (a) is split in three lines by means of MW signal splitters (b). It passes through the circulators (c), reaching the TTL controlled switches (d), which are powered by a common power line (f). The phase shifters (e) shift the signals, which are subsequently amplified (g) and attenuated by continuous variable attenuators tuned by micrometer screws (h). Finally, the signals are output (i) to the MW antennas, which are placed inside the vacuum chamber.

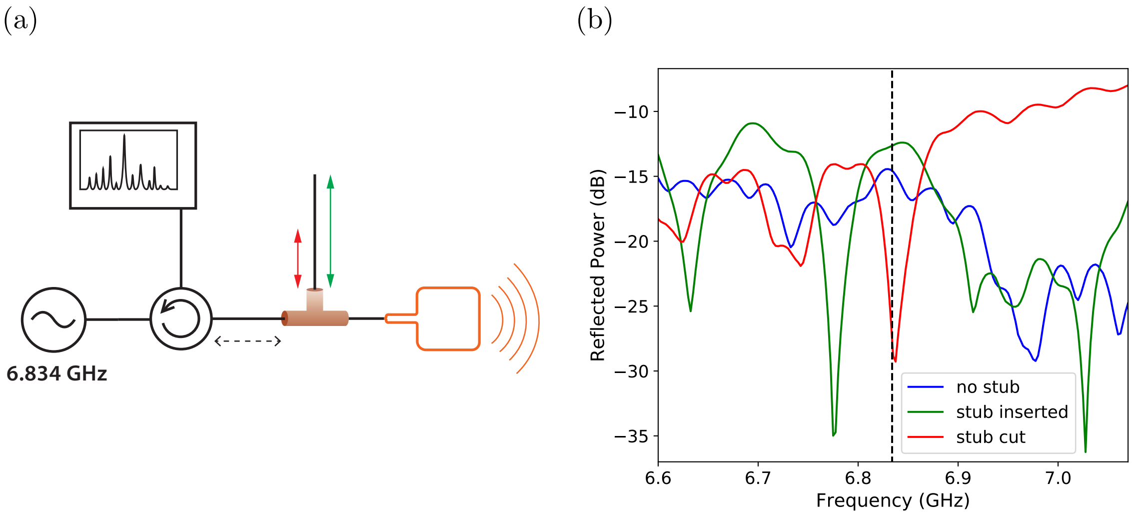


Fig. 4.5.: (a) Sketch of the setup used for the stub tuning method. The stub is placed via a T-connector - colored in brown - between the circulator and the antenna and the spectrum analyzer measures the power reflected by the system composed by stub and antenna. The stub initial length is sketched in green, whereas its final length, reached after cutting it multiple times, is sketched in red: these colors correspond to the plot lines in the figure on the right. For a perfect impedance matching, also the stub insertion point, sketched by the horizontal dashed arrow, should be modified.(b) Reflected power measured by the spectrum analyzer as a function of frequency. The vertical dashed line indicates 6.834 GHz, where the reflection is reduced of 12 dB - from −15 dB (blue line) to −27 dB (red line).

$$C(\tau) = \int_{t_0}^{t_1} \mathrm{pulse}_1(t)\mathrm{pulse}_2(t+\tau)dt \tag{4.1}$$

and its maximum point provides the temporal delay between the two pulses. The computed cross-correlation is plotted in Fig. 4.7b and it results in a delay at most of 15 ns, which is totally negligible.

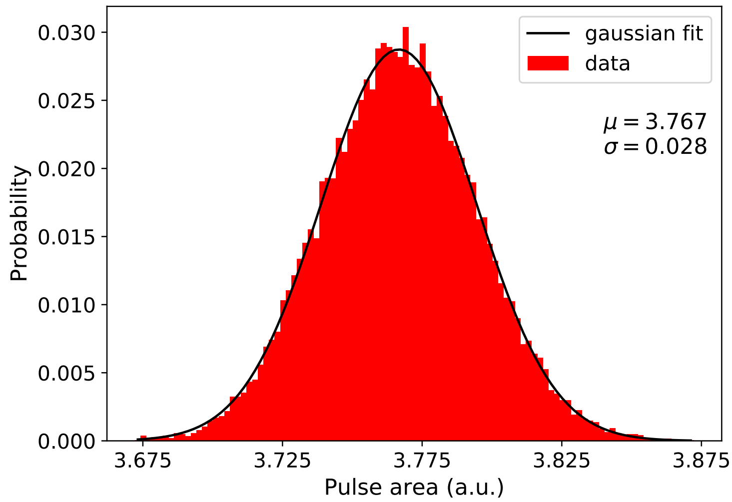


Fig. 4.6.: Power stability histogram. It shows the measured occurrences as a function of detected pulse area in arbitrary units, fitted by a Gaussian function with relative error $\sigma/\mu = 0.7\%$.

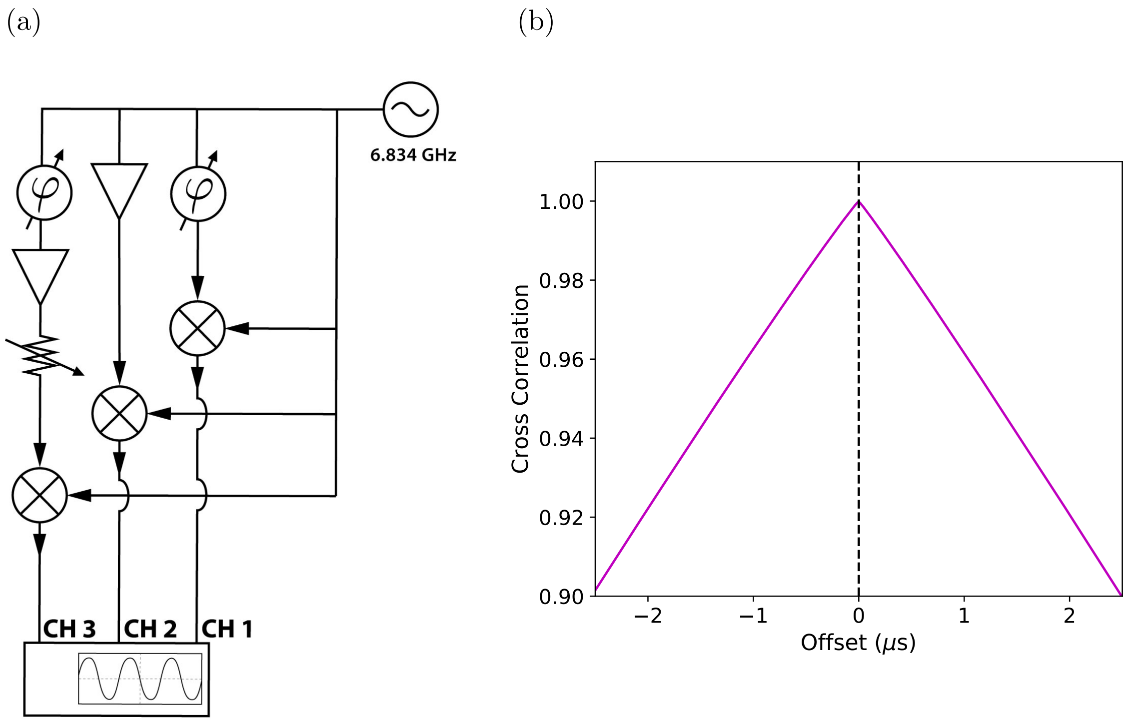


Fig. 4.7.: (a) The MW pulses are downmixed with the same frequency and the resulting rectangular pulses are measured in the oscilloscope. The MW signals are sent through three different electrical paths - the first only with phase shifter, the second only with amplifier and the third with phase shifter, amplifier and attenuator - to test the pulses retardation introduced by the elements. (b) The downmixed rectangular pulses are detected in the oscilloscope and cross-correlated via eq. (4.1), resulting in a retardation at most of 15 ns. The normalized cross correlation is plotted as a function of the offset time τ and the vertical dashed line indicates its maximum point.

4.3. Measuring the polarization with single atoms

4.3.1. Experimental protocol

The protocol steps to control the MW field polarization described in section 3.2 are absolutely general and independent of whether MW fields are measured with dipolar antennas - as for the test apparatus - or with atoms. Nevertheless, since the receiving antennas are replaced by a single atom, the MW field information has to be extracted by different measurements.
The 3D Jones vector (2.22) is completely characterized by three amplitudes and two relative phases which are all related to the Rabi frequencies of the different hyperfine-ground-state transitions, as derived in section 2.4.3: thus the measurement principle boils down to Rabi frequencies measurements.
As first step, the quantization axis is provided by a magnetic field of $\sim$ 470 mG aligned along the long cavity direction, called z by convention, which additionally gives the hyperfine Zeeman splitting. The atom is prepared into $F = 1$, $m_F = 0$ via Zeeman pumping and a MW spectrum over the hyperfine-ground-state transitions is measured. Figure 4.9 shows a MW spectrum in which only three peaks are visible, indicating a successful preparation of the atom in the desired state, with a contribution to the background of roughly 1%. By fitting the peaks with the function (2.10) it is possible to derive the Rabi frequencies $\Omega_{|1,0\rangle\to|2,0\rangle,\,z}$, $\Omega_{|1,0\rangle\to|2,1\rangle,\,z}$ and $\Omega_{|1,0\rangle\to|2,-1\rangle,\,z}$, which are related to magnetic field's polarization through the equations (2.43 - 2.45).

As equations (2.51-2.53) suggest, the second step could consist in repeating the same procedure, this time employing a different guiding magnetic field to provide an orthogonal quantization axis. For this purpose, the magnetic field was aligned along the short cavity direction, called y by convenience. Nevertheless, as soon as the quantization axis is changed, the experimental setup requires technical changes and adjustments to successfully pump the atom in the Zeeman state $F = 1$, $m_F = 0$, being actually able only to pump in the manifold $F = 1$. This additional feature makes the spectra no longer suitable to infer directly Rabi frequencies, due to the population spread among $F = 1$ states.
By driving Rabi oscillations on $F = 1$, $m_F = 0 \longleftrightarrow F = 2$, $m_F = 0$ and by fitting the result with eq. (2.10), assuming $\Delta = 0$ and considering the initial population as an additional fit parameter, is possible to obtain $\Omega_{|1,0\rangle\to|2,0\rangle,\,y}$. The same measurement is repeated on the edge transitions $F = 1$, $m_F = 1 \longleftrightarrow F = 2$, $m_F = 2$ or on $F = 1$, $m_F = -1 \longleftrightarrow F = 2$, $m_F = -2$, being the only transitions which are coupled, respectively, solely by σ^+ and σ^- contributions to the polarization. Therefore the polarization is derived from the relations

$$\Omega_{|1,0\rangle\to|2,0\rangle,\,z} = \chi B_{0z}, \tag{4.2}$$

$$\Omega_{|1,0\rangle\to|2,1\rangle,\,z} = \chi\sqrt{\frac{3}{8}}\sqrt{(B_{0x} - B_{0y}\sin\delta_1)^2 + B_{0y}^2\cos^2\delta_1}, \tag{4.3}$$

$$\Omega_{|1,0\rangle\to|2,-1\rangle,\,z} = \chi\sqrt{\frac{3}{8}}\sqrt{(B_{0x} + B_{0y}\sin\delta_1)^2 + B_{0y}^2\cos^2\delta_1}, \tag{4.4}$$

$$\Omega_{|1,0\rangle \to |2,0\rangle,\, y} = \chi B_{0y}, \tag{4.5}$$

$$\Omega_{|1,1\rangle \to |2,2\rangle,\, y} = \chi \sqrt{\frac{3}{4}} \sqrt{\left(B_{0x} + B_{0z} \sin \delta_2\right)^2 + B_{0z}^2 \cos^2 \delta_2}, \tag{4.6}$$

$$\Omega_{|1,-1\rangle \to |2,-2\rangle,\, y} = \chi \sqrt{\frac{3}{4}} \sqrt{\left(B_{0x} - B_{0z} \sin \delta_2\right)^2 + B_{0z}^2 \cos^2 \delta_2}, \tag{4.7}$$

where $\chi = \frac{\mu_B}{2\hbar}(g_S - g_I) = 8.8 \times 10^{-10}\,\mathrm{s\,T^{-1}}$ is a factor. The last two equations (4.6-4.7) are obtained by repeating the calculations described in section 2.4.2 for the transitions $|1,1\rangle \longleftrightarrow |2,2\rangle$ and $|1,-1\rangle \longleftrightarrow |2,-2\rangle$.
Since the trigonometrical equations provide two different solutions for each phase - namely δ and $\pi - \delta$ - a further measurement along a third orthogonal axis is performed to discriminate the correct solution among these.
Figure 4.10 shows representative plots for Rabi oscillation measurements both on the clock and on the edge transitions; the latter underlies strong decoherence, preventing the observation of Rabi oscillations. This is due to the fact that these transitions involve states highly susceptible to magnetic field fluctuations, since the energy Zeeman shift of these states scales with the magnetic quantum number m_F. This results in a statistical relative error up to 15 % on the Rabi frequency produced by the fits.
The effect of decoherence effects is to damp the oscillations and can be shown that the probability of transition to $F = 2$ states $P_{|2\rangle}(t)$ is given by the following expression [36]:

$$P_{|2\rangle}(t) = p_1 \frac{1}{2(1+2\xi^2)} \left[1 - \left(\cos(wt) + \frac{3\xi}{\sqrt{4-\xi^2}} \sin(wt) \right) e^{-\frac{3}{2}\gamma t} \right], \tag{4.8}$$

where γ is the damping rate and the parameters ξ and w are

$$w = \Omega_R \sqrt{1 - \frac{\xi^2}{4}} \tag{4.9}$$

$$\xi = \frac{\gamma}{\Omega_R} \tag{4.10}$$

It is simple to verify that in the strong-field regime, i.e. with the condition $\Omega_R \gg \gamma$, eq. (4.8) boils down to equation (2.10).

The full sequence for atom preparation and measurement is presented in fig. 4.8, showing that the time scale of the experiment is dominated by the cooling sequence time. The reason for this is that the MW pulses deposit their energy in the vacuum chamber and, if they are too frequent or too strong, could damage the cavity setup. The MW pulses become less frequent if the experiment total duration is increased.

Thus the polarization measurement protocol, consisting of one MW spectrum (quantization axis along z) and three Rabi oscillation measurements (three with quantization axis along y, whereas the extra third one with quantization axis along x), is repeated for every antenna independently and for all three of them together. It requires, therefore, a total number of $4 \times 4 = 16$ measurements to completely characterize the 3D polarization of the magnetic field, after which phase shifters and attenuators can be adapted in order to obtain the aimed polarization.

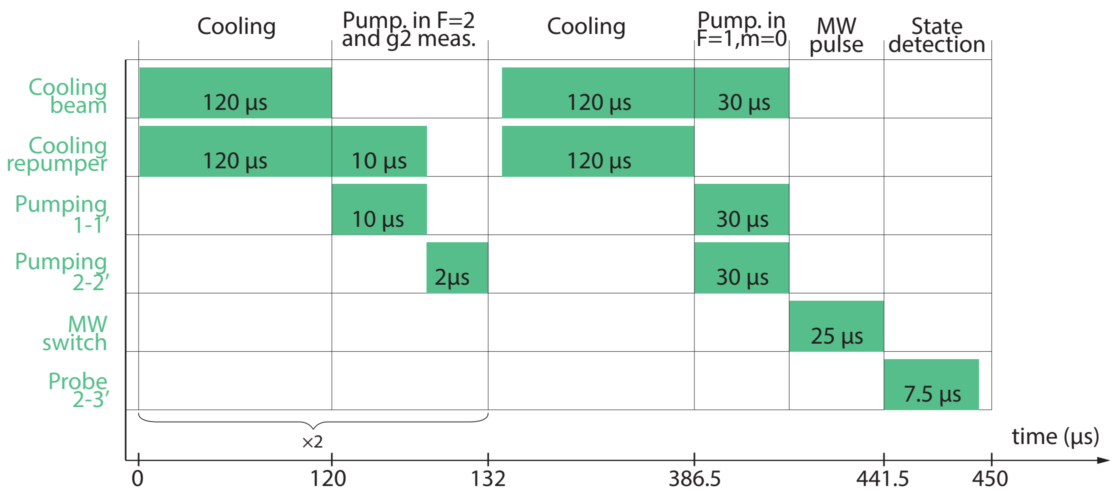


Fig. 4.8.: Sequence scheme used to perform MW spectroscopy. A full sequence takes 450 µs and is thus repeated with a rate of 2.2 kHz while the MW frequency is swept over a range of ±1.5 MHz around 6.834 626 GHz in almost 2 s. As explained in the main text, the time scale of the experiment is dominated by the cooling time, which takes 80 % of the total duration.
The fast sequence used for Rabi oscillation measurements is very similar. The subsequence "Cooling + $g^{(2)}$ measurement" is repeated 3 times, instead of 2, and the MW frequency is held constant, whereas the pulse duration time is swept from 0 µs to 120 µs. A full sequence takes therefore 557 - 677 µs. The sequence segments drawn in this picture are not in scale.

4.3.2. Polarization selection results

After a complete characterization of the magnetic field 3D polarization, phase shifters and attenuators were adjusted to obtain the desired polarization and MW spectra were measured to check that the peaks corresponding to the undesired polarizations were suppressed. In principle, the resulting polarization could have been fully measured - as was done for the single antennas - and quantitatively compared to the expected one as was done in section 3.3 via the overlap between the theoretical and measured Jones vectors, defined in eq. (3.4).
Nevertheless, due to the appearance of an issue in the experimental setup - see the Appendix A - it has been possible to measure only a few MW spectra, neither allowing a complete quantitative data analysis, nor a subsequent improvement of the results.
Figure 4.11 reports the resulting MW spectra for different aimed polarizations, namely for π, σ^- and σ^+ along the y-axis. In the spectra for π and σ^- polarization, the partial suppression of the undesired polarizations can be observed. In order to quantify these result, the Rabi frequencies suppression factor f is defined. For example, if the suppression of σ^- for aimed π-polarization is considered, f is given by

$$f = \frac{\Omega'_\pi}{\Omega'_{\sigma^-}} / \frac{\Omega_\pi}{\Omega_{\sigma^-}}, \tag{4.11}$$

where with Ω' and Ω are indicated the Rabi frequencies after and before the polarization manipulation, respectively. This means that the suppression factor f is defined as the ratio between the Rabi frequencies of the aimed polarization and of the suppressed one, relatively to the initial polarization.

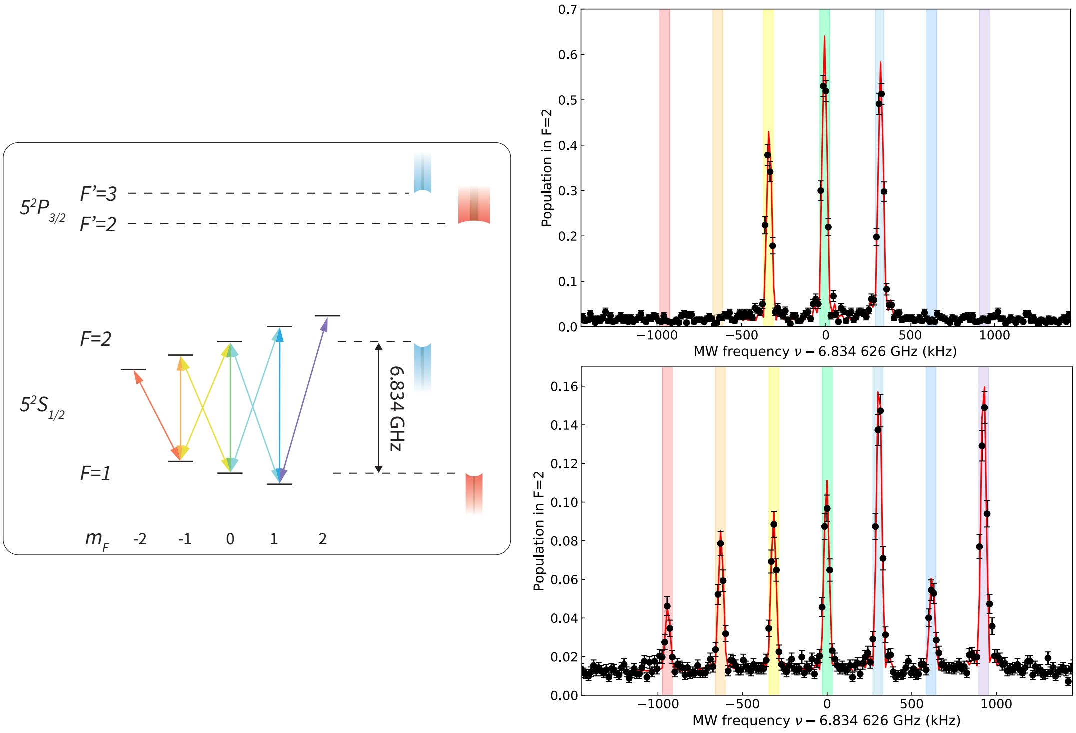


Fig. 4.9.: The MW spectroscopy is performed on the atom in order to derive the transition frequencies. The atom is initialized in the manifold $F = 1$ and the population transferred to $F = 2$ by MW pulses is measured by the state detection with the short cavity tuned on $2 - 3'$, as depicted in the left box. The plot on the upper right shows a MW spectrum obtained by driving only one antenna after the optical pumping into Zeeman state $F = 1$, $m_F = 0$, since only three central peaks are visible, with a background over all the spectrum of less than 2%. For this measurement, the guiding magnetic field is aligned along z-direction. The colored windows in the plots highlight which transitions the peaks correspond to, with reference to the left box. The peaks are fitted with eq. (2.10) as function of the detuning. The plot in the lower right shows a MW spectrum for the same antenna and guiding magnetic field along y-direction, indicating hyperfine pumping into the whole $F = 1$ manifold. The error bars correspond to one standard deviation.

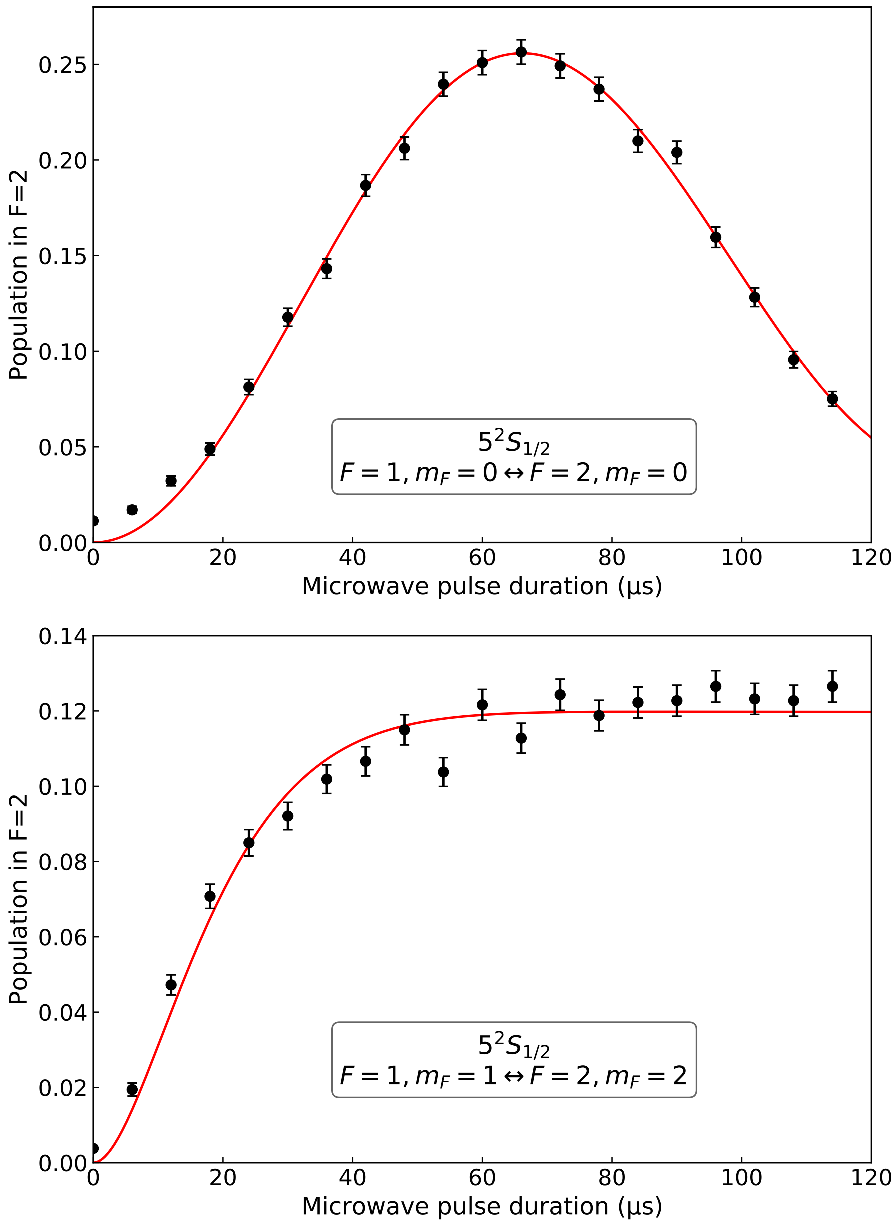


Fig. 4.10.: The microwaves are tuned resonantly to the clock transition $F = 1$, $m_F = 0 \longleftrightarrow F = 2$, $m_F = 0$ and on the edge transition $F = 1$, $m_F = 1 \longleftrightarrow F = 2$, $m_F = 2$ and Rabi oscillations measurements are performed on the atom in order to derive the corresponding Rabi frequencies. After being prepared in the whole manifold $F = 1$, MW pulses of increasing duration are applied on the atom, and the population transferred to $F = 2$ is measured by the state detection on $2 - 3'$. The upper plot shows a Rabi oscillation performed by feeding only one antenna on the clock transition, resulting in a Rabi frequency of $\Omega_R = 2\pi \times 7.4\,\text{kHz}\pm\ 0.5\%$, a decay rate of $\gamma = 2\pi \times 0.3\,\text{kHz} \pm 22\%$ and an initial population $p_1 = 26 \pm 1\,\%$, obtained by fitting the points with eq. (4.8). The resulting best fit is plotted in red. The lower plot shows a coherent Rabi driving performed with the same antenna on the edge transition $m_F = 1 \longleftrightarrow m_F = 2$, clearly indicating a weak-field regime, in which decoherence effects dominate the driving, i.e. the condition $\Omega_R \ll \gamma$. The fit through eq. (4.8) provides a Rabi frequency of $\Omega_R = 2\pi \times 9.7\,\text{kHz} \pm 6\%$ and a decay rate of $\gamma = 2\pi \times 6.8\,\text{kHz} \pm 7\%$. The error bars correspond to one standard deviation.

Even though the Rabi frequencies could not be extracted from the MW spectra, it is possible to calculate the the suppression factor f. Indeed, if p_{max} is a peak maximum, from (2.10) it results $\Omega \propto \arcsin\sqrt{p_{max}}$, thus boiling down expression (4.11) to

$$f = \frac{\arcsin\sqrt{p'_{max,\pi}}}{\arcsin\sqrt{p'_{max,\sigma^-}}} \Bigg/ \frac{\arcsin\sqrt{p_{max,\pi}}}{\arcsin\sqrt{p_{max,\sigma^-}}}. \tag{4.12}$$

Here, p'_{max} and p_{max} indicate the transition peak maximum after and before polarization manipulation, respectively.
The π-spectrum returns a Rabi frequencies suppression factor for σ^- and σ^+ respectively of 2.1 ± 0.2 and 1.4 ± 0.1. The σ^--spectrum returns a suppression of factor 1.4 ± 0.1 for σ^+ and shows no transitions that are coupled by a π-polarized field. Thus both of them follow, at least qualitatively, the predictions. In particular, the absence of transitions coupled by π-polarized field in the σ^--spectrum is a positive indication for the achievement of the purpose.
The σ^+-spectrum instead shows a strong suppression of σ^+ polarization, deviating from the prediction. This is mainly due to three reasons:

- In the first place, a mistake in the code used to derive the polarization from the measured Rabi frequencies was detected and, unfortunately, was not possible to repeat the measurements after having corrected it;

- even without code mistakes, the following two are the limiting factors to this experiment. Due to the strong decoherence effects, the Rabi driving on the edge transitions return Rabi frequencies which have a considerable error and strongly depend on the fitting parameters. In particular, if the population initialized in the involved Zeeman state of $F = 1$ and inserted in the model as a free parameter can range over a considerable portion of $(0, 1)$ without affecting the fit quality, the fitted Rabi frequency will range in turn;

- since the amplitude of oscillation is determined by the initial population, the maximum amplitude of Rabi oscillations differs from 1, making impossible to test, from a single Rabi oscillation measurement, whether the microwaves are tuned precisely on resonance, introducing a further uncertainty on the Rabi frequencies measured with this method.

A strategy to overcome these limiting factors consists of being able to perform ground state Zeeman pumping in $F = 1, m_F = 0$ also along the y-quantization axis. This allows measuring the Rabi frequencies directly by fitting the MW spectrum peaks, thus avoiding to take Rabi oscillations measurements.
While the π-pumper along the z-axis - which is the key to accumulate population exclusively into $m_F = 0$ - is provided by a vertical beam, a π-pumper along the vertical axis, perpendicular to the cavity plane, can be provided by an intracavity beam. By using this axis as y-axis, the protocol can be performed by measuring only MW spectra along the z- and y-axes.
Generally, when it comes to light propagating through fibers, the fiber polarization maintenance has to be taken into account if a precise light polarization has to be provided. Although this may be an issue - since the light is coupled into the cavities by fibers - the birefringence of the short cavity can be used to select the transmitted polarized light.

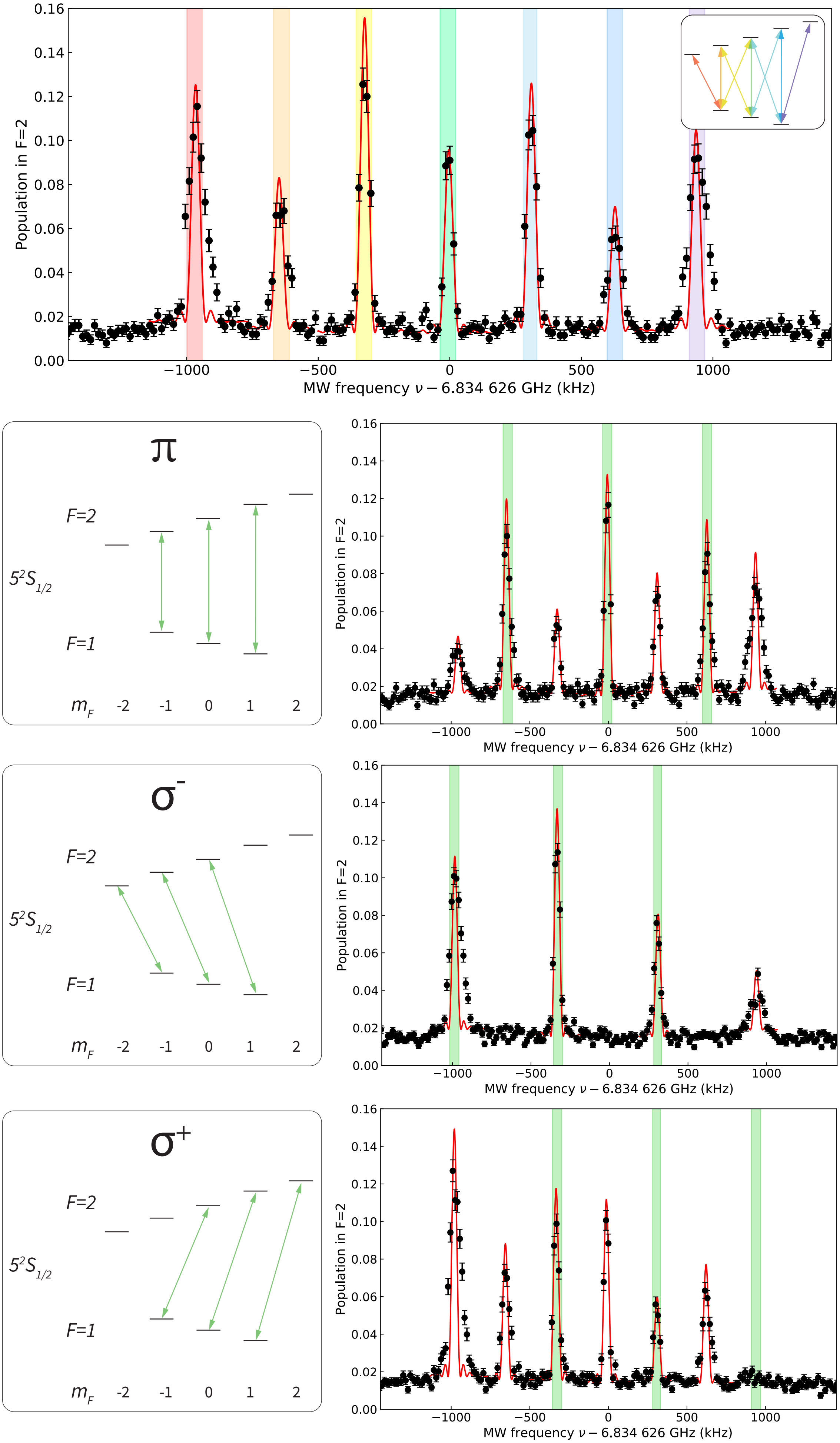

Population in F=2
MW frequency ν − 6.834 626 GHz (kHz)
π
σ-
σ+
F=2
F=1
$5^2S_{1/2}$
m_F -2 -1 0 1 2

Fig. 4.11.: MW spectra resulting from the protocol described. All the spectra reported are measured with all three antennas simultaneously while the guiding magnetic field is set along the y-axis.
In order to estimate the performance of the method, the spectrum taken before the insertion of phase shifts and attenuations is plotted on the top. All the peaks are fitted in red by the function (2.10) multiplied by the population in the $F = 1$ corresponding state, entering the model as an additional parameter. The error bars correspond to one standard deviation. The small inset clarifies the correspondence between peaks and hyperfine transitions.
The resulting MW spectra for π, σ^- and σ^+ polarization are shown; the green windows highlight the peaks corresponding to the transitions coupled by the field with the aimed polarization component, revealing an imperfect achievement of the purpose. For π-polarization, σ^- and σ^+ Rabi frequencies are suppressed respectively of a factor 2.1 ± 0.2 and 1.4 ± 0.1 . For σ^--polarization, σ^+ is suppressed of a factor 1.4 ± 0.1 and peaks corresponding to transitions coupled by π-polarized field are not visible.
Additionally is possible to notice that the edge transitions peaks are broadened as a consequence of decoherence effects, making the above-mentioned fitting function no longer suitable.

By employing elliptical mirrors in building a cavity, i.e. mirrors that present two different radii of curvature along two principal axes, is possible to lift the degeneracy of the polarization eigenmodes, namely splitting the transmission frequencies of these modes [37]. In this case, the short cavity out-coupling mirror has radii of curvature 100 µm and 290 µm, whereas the high-reflecting mirror has radii of 90 µm and 230 µm, resulting in a polarization eigenmodes splitting of four linewidths [31].

5. Conclusion

This project was aimed at the use of polarization controlled microwave (MW) fields in order to manipulate the electronic ground state of a single atom coupled to high-finesse cavities. To select the MW polarization in a three-dimensional space, the MW signal feeding three single-loop antennas was controlled in phase and amplitude, thus manipulating the interference produced by the MW fields. The methods adopted were theoretically and experimentally explored, both with a test apparatus, separated from the cavities and outside the vacuum chamber, and by means of a single atom located at the crossing point of the independent cavity modes.
In the former, the microwaves were detected by three orthogonal dipolar antennas where the output signal was displayed and analyzed by an oscilloscope. The MW test setup assembly was detailed and characterized. It showed a good agreement of theoretically expected and measured polarizations, resulting in a mean polarization fidelity of $(98.9 \pm 0.5)\%$ and thus confirmed the validity of the polarization manipulation strategy.
In the vacuum chamber, single ^{87}Rb atoms probed the magnetic field polarization. The experimental setup consisted of two crossed fiber cavities in which single ^{87}Rb atoms were loaded and, after state preparation, were impinged by rectangularly shaped MW pulses, undergoing $5S_{1/2}$ hyperfine ground state transitions between $F = 1$ and $F = 2$. The polarization measurement was enabled by the measurement of ground state transition Rabi frequencies through microwave spectra and measurements of Rabi oscillations along different quantization axes. The experimental results obtained with the atom showed a measured polarization only qualitatively and partially following the predictions, thus revealing systematic errors. The main imperfections and limiting factors were discussed. In particular, a significant factor of uncertainty affects the measurement of the Rabi frequencies derived by Rabi oscillations measurements, due to the comparably high decoherence rate. This triggers the second uncertainty, namely, the resonant tuning of the MW field to the atomic transitions.
Another factor affecting the presented results, was the appearance of an issue in the experimental setup involving one of the cavities, impeding further avenues of improvement. Related problems and strategies for recovering the ready-to-work status are described in the Appendix.
Even though the deterministic selection of MW polarization was not achieved with the atom, it does present a significant room for improvement. A promising strategy, consisting of performing Zeeman pumping into the state $5S_{1/2}, F = 1, m_F = 0$ along two orthogonal quantization axes would provide a more reliable measure of the Rabi frequencies. A practicable way to achieve this with the employed cavity setup was proposed. For future projects, the employment of more compact and less dispersive microwave sources, such as microwave microstrips [19], placed closer to the atom could be explored as an alternative to the antennas in order to provide more power and to limit the risk of damage to the apparatus.
Overall, this work presents a contribution to the achievement of coherent state ma-

nipulation of atomic ground states, whose energy splitting lies in the microwave domain. Since such states very often play the role of qubit-states [32, 38, 39], this provides a further tool to the quantum information processing field, enabling, for example, a microwave population transfer in decoherence-protected states [14], dynamical decoupling schemes or fast atomic state transfer.

A. Short Cavity Issues

For purposes of cavity stabilization and monitoring, light reflected and transmitted by both cavities is constantly detected by photodiodes. During the time this project was conducted, from one day to another neither reflected nor transmitted light via the short cavity could be anymore detected, whereas the long cavity showed a usual correct functioning. On the same day, a vacuum-chamber pressure of 9×10^{-10} mbar was measured, significantly larger than the usual pressure value of 2.1×10^{-10} mbar - see section A.2. These observations suggested initially damage of the fiber cavity, which eventually led to a release of matter in the chamber with a subsequent increase of pressure.
The issues are treated separately, the strategies used to tackle them are described and the results are discussed.

A.1. Absence of light

The absence of light reflected and transmitted by the short fiber cavity is compatible with two different hypotheses: on one hand, a damage of the cavity could cause scattering of the light out of the cavity mode and of the fiber itself, on the other hand, a bad mode matching of the light into the single-mode fiber would result mainly into scattering on the fiber tip instead of correct incoupling into the fiber.
In order to check the status of the cavities, they were imaged with an electron multiplying charge-coupled device (EMCCD) camera while 780 nm light of 100 nW was shined into the SM fibers and both cavities were out of resonance. The imaging system employed is described in the doctoral thesis of Dominik Niemietz [40].

The result is shown in Figure A.1a and is interpreted in the following way. The light in the long cavity fiber is partially reflected by the out-coupling mirror out of the supported mode, escaping the fiber and being imaged by the Camera; the short cavity is instead almost invisible, even though the maximum gain was used as opposed to the long cavity, showing only a few faint illuminated spots. This observation seems to indicate the absence of light incoupled into the fiber, supporting the hypothesis of bad fiber coupling.
By looking through an infrared (IR) viewer was not possible to distinguish any clear spot on the single mode (SM) fiber tip where the beam is incoupled from the laser into the fiber. The use of charge-coupled device (CCD) cameras pointing to both short and long cavity fibers helped in detecting light scattered from the fibers and the holders; nevertheless, the comparison between the two did not help further in concluding the cause of the issue.
Even though the CCD cameras did not provide conclusive answers, they were helpful in the alignment of an auxiliary beam along the same optical path of the in-coupled beam, so that it was possible, to some extent, to handle the fiber-coupling of the auxiliary beam without altering the in-coupled one. After improving the focus of the short cavity in the EMCCD camera, it was possible to detect some light spots on the

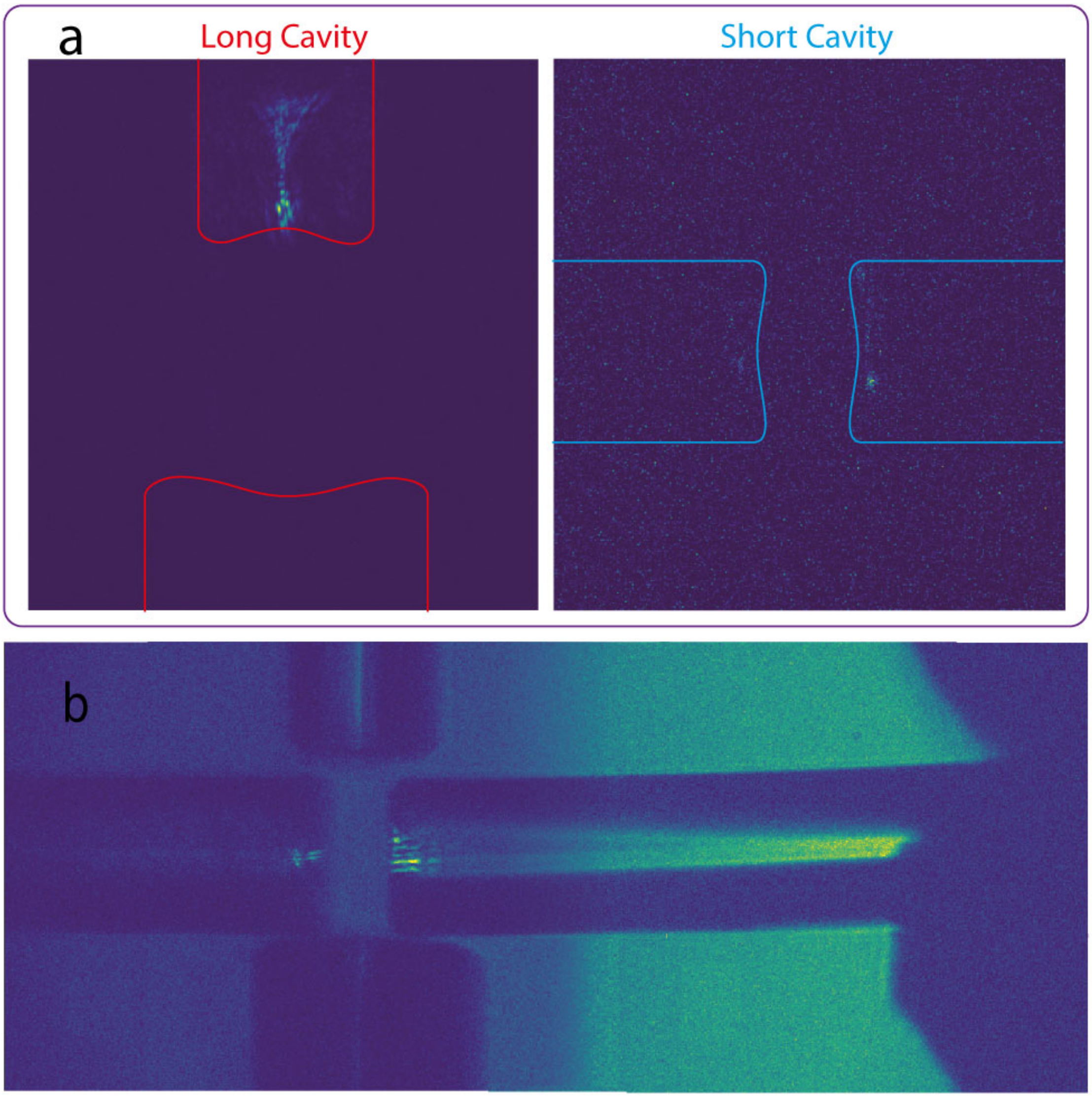


Fig. A.1.: Fiber cavities imaged with an EMCCD camera. (a) Both cavities are imaged separately while 780 nm light at 100 nW is shined through the SM fibers. The cavity shapes are drawn for visualization purposes. For the long cavity, no camera gain was employed, and is possible to distinguish the light reflected by the out-coupling mirror and scattered by the cladding. For the short cavity, the maximum gain was employed, allowing to see only a few fainted illuminated spots. (b) Further segments of the short cavity fiber. By keeping the camera focused on the short cavity fiber and shining 780 nm light through the SM short cavity fiber and additional 780 nm light from the bottom of the chamber, it is possible to image the short cavity and some further fiber segments. It is moreover possible to distinguish several illuminated spots on both out-coupling and high-reflecting mirrors.

mirrors, as shown in Figure A.1b. Their visibility was optimized through iterative alignments of the auxiliary beam, yielding the detection of transmitted light via an SPCM by scanning repeatedly the cavity length. Figure A.2 reports the resonance peak seen in the transmission channel and proves that an actual cavity signal was measured. Further alignment optimizations were performed by maximizing this peak.

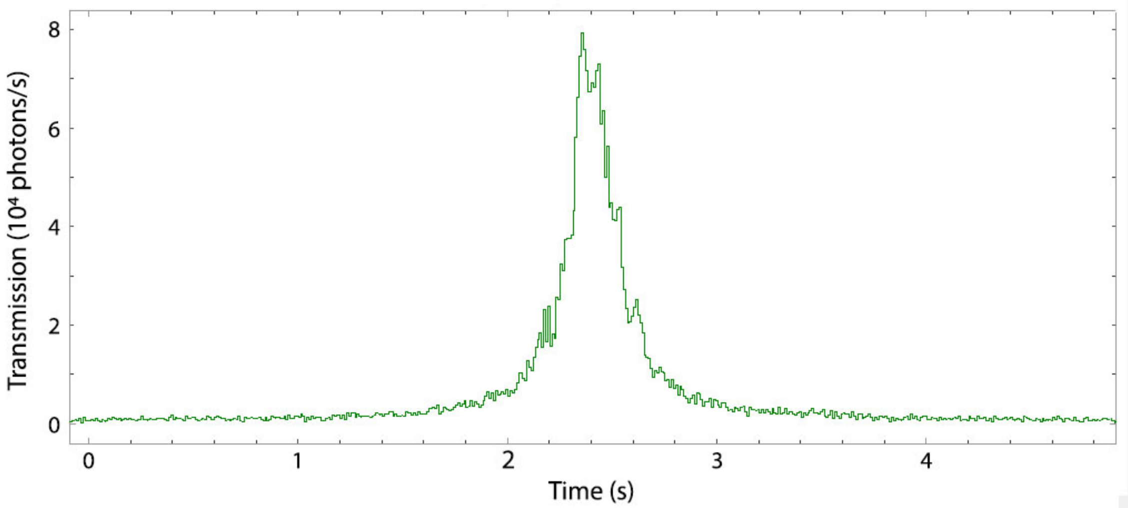


Fig. A.2.: The light transmitted by the short cavity is detected by an SPCM and the cavity-length is scanned in time, resulting in a transmission peak of almost 10^5 photons/s.

Although the inferred linewidth is comparable with its usual value, thus indicating intact cavity mirrors, the measured signal is remarkably smaller than expected - 10^5 photons/s measured against 10^9 photons/s expected in transmission. Further segments of the short cavity fiber inside the chamber were imaged and are shown in Figure A.1b.

A.2. Vacuum chamber pressure

The same day the absence of light in the short cavity emerged, an increase in the vacuum-chamber pressure was noticed and since then, linear growth of the pressure was monitored, plotted in Fig. A.3. The pressure was last checked some days earlier, making it hard to estimate when it started to rise - the value was settled on 2.1×10^{-10} mbar for a time > 1.5 years, whereas a value of 9×10^{-10} mbar was measured.

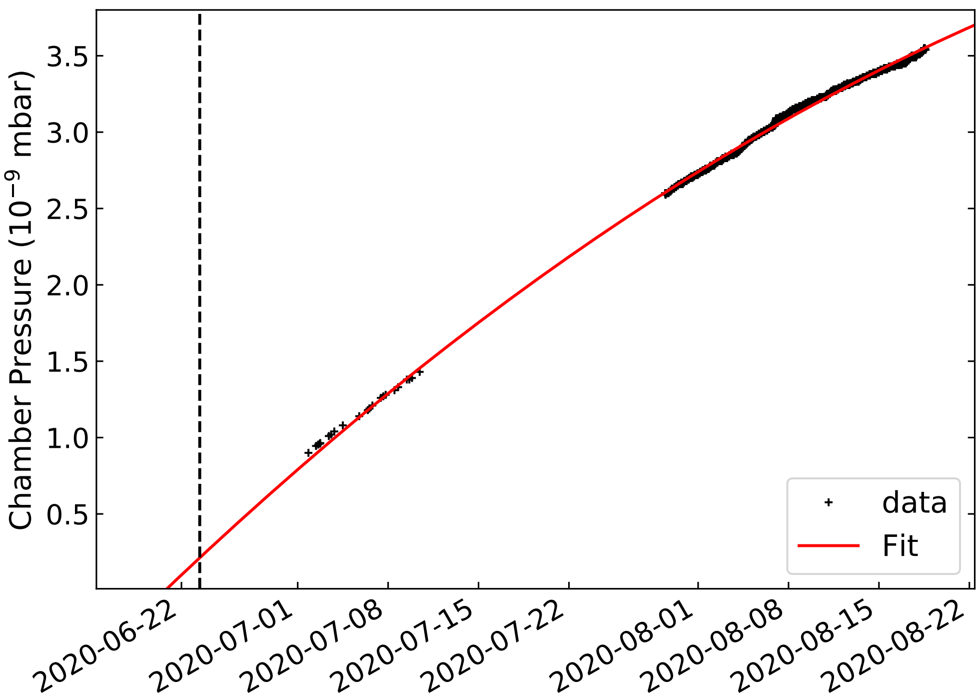


Fig. A.3.: The pressure growth is monitored over two months since the day the issues were detected. The data are fitted with a polynomial function, showing a rate slightly decreasing over time - the fit curve bends a little downwards. The vertical dashed line indicates the day the pressure started increasing from the usual value of 2.1×10^{-10} mbar, according to the fit.

This observation is compatible with a leakage in the vacuum, with a release of material from the fiber due to damage - producing a result similar to a leakage in the vacuum - and with a not fully working vacuum pump system. The vacuum system employed (*SAES GROUP NEXTORR D500-STARCELL*)is a combined system consisting of a passive porous getter, which pumps out active gases at room temperature without the need for electric power to operate, featuring a pumping speed of $500\,\mathrm{L\,s^{-1}}(H_2)$, and of an ion pump featuring up to $30\,\mathrm{L\,s^{-1}}(CH_4)$. The overall performance is such to keep UHV in the vacuum chamber used, that has 10 L of

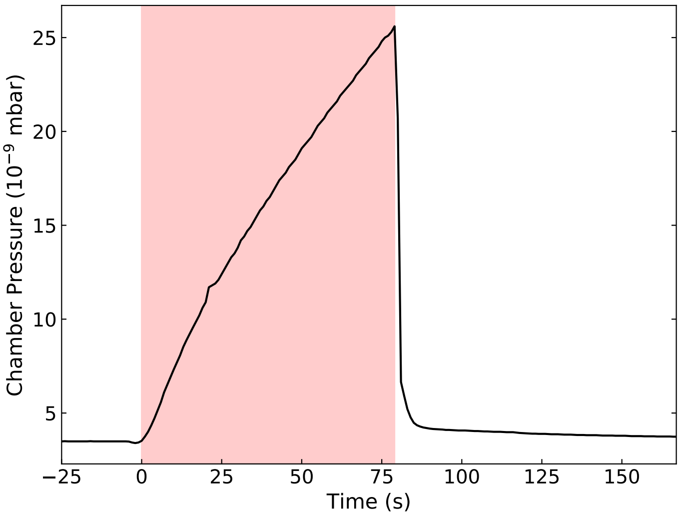


Fig. A.4.: The pressure growth is monitored while the ion pump is switched off. The red window indicates the time interval in which the pump was off, for a total time of roughly 1:20 minutes. The increasing rate is much higher than the slow growth monitored over two months - after only 25 s the pressure is increased of an order of magnitude. As soon as the ion pump is reactivated, the pressure decreases instantaneously to its initial value.

volume; further details can be found in [31].
Since the non-evaporative getter (NEG) pump is a passive absorbing pump, it can saturate completely and thus require a reactivation. A preliminary test was performed, consisting of monitoring the pressure for a short time while the ion pump was switched off. The result, shown in Fig. A.4, is compatible both with a leak, as there could be a non-negligible amount of non-getterable gases (Ar or CH_4) not pumped by the NEG alone and as soon as the pumping contribution of the ion pump is removed, the pressure rises, or with a fully saturated NEG, meaning that the ion pump is the only pumping the system.
Repeating the same test after having reactivated the NEG could help to determine the presence of a leak.

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Declaration

I hereby declare that this thesis is my own work, and that I have not used any sources and aids other than those stated in the thesis.

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Author's signature